

DFO-LEP Acoustic Target Data Issues (1977-2007).pdf

Read_Me

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READ_ME

Project Name: DFO-LEP Acoustic Target Data (1977-2007)

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Date file was created: June 10, 2025

Date file was finalized: January, 2026

The Datasets

These acoustic target data were collected as part of DFO's Lake Enrichment Program (LEP). See Hyatt et al. (2000), Hyatt et al. (1984) and Gjernes et al. (1986) for background details on acoustic limnetic fish census methods employed to obtain fish counts for the LEP. Ideally, the acoustic target data should be considered together with the trawl biosampling data. See references section.

Data Collection

Data were collected in freshwater lakes located in British Columbia, Canada, from 1977 until 2007 (and a few surveys in early 2008 that were part of the cohort that was sampled in 2007). Data were collected using hydro-acoustic counts from echo-sounders of "fish-like targets" (FLT). The echosounder data were originally processed by lab technicians using proprietary software associated with the echosounders, and saved into annual datafiles titled TARGETyy.DAT, where yy indicates a two-digit value representing sampling years from 1977 to 2007, and refers to the Sockeye juvenile salmon cohort year. The cohort year spans an April – March timeframe, from 01-Apr-yy to 31-Mar-yy+1, and is also referred to as the acoustic-trawl-survey year, or "ATS year" (details below; see section

Target_CHK_exact_duplicate_records – list of records removed due to duplication of data across all fields or columns

- **Target_CHK_keyfield_duplicate_surveys** – list of surveys duplicated on key fields (lake, survey_date, survey_type, sounder_type)
- **Target_CHK_sequential_surveys** – list of surveys that occur within one day of each other for the same lake, representing possible replicate surveys (change in field survey method (e.g., sounder type), or lab analysis method (e.g., software used, technician performed)).

- Target_CHK_targets_issues – list of remaining records (if any) with anomalous data issues
- Target_CHK_RAW_vs_CLEAN_surveys – inventory of records organized by whether they exist only in TARGETty.DAT files (i.e., omitted from cleaned dataset), only in cleaned data (programmatic additions or key-field revisions), or in both.

The contents of these files are further described in “DFO-LEP Acoustic Target Data Issues (1977-2007).pdf”.

Note that the **ACOUSTIC TARGET DATA CLEANUP LOG.txt** file shown above contains the RStudio console output of a successful run of the R program (not automatically saved to the \output folder).

Output Data Review

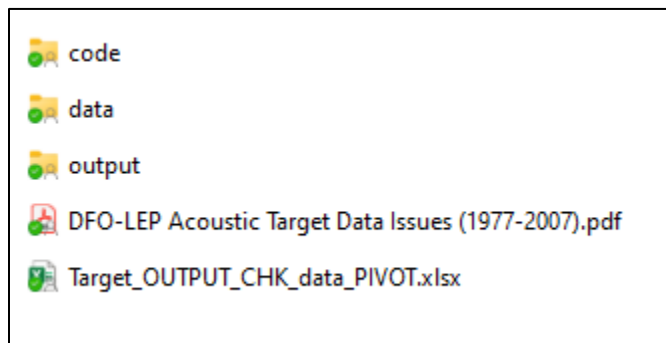
While the CSV files can each be opened in MS-Excel or read into R for review and further analysis, a quick look at the R program’s output files can be obtained with the included Excel workbook:

“Target_OUTPUT_CHK_data_PIVOT.xlsx”. This workbook contains multiple worksheets, each automatically linked to a corresponding CSV file for ease of data display, filtering, and pivot summary.

Once initial setup is performed (see details in section “[To Use the Target Output Checking Data Workbook](#)” in “DFO-LEP Acoustic Target Data Issues (1977-2007).pdf”), the data in “Target_OUTPUT_CHK_data_PIVOT.xlsx” worksheets are automatically updated with the contents of the CSVs upon opening the workbook or refreshing the data.

Additional Metadata Included as a ZIP File

The zip file, **DFO-LEP Acoustic Target Data (1977-2007).zip**, contains the following folders, following the subdirectory structure of the RStudio project employed to run the Acoustic Target Data Cleanup.R program:



Descriptions of Variables). The TARGETty.DAT files are essentially text files and are readable by a text editor such as Notepad.

Data Cleaning

The TARGETty.DAT files are imported into data-cleaning software developed by a team comprising Yuliya Shtymburska (Canadian Institute of Ecology and Evolution), Howard Stiff (DFO), Athena Ogden (DFO), and Sandra Emry (Canadian Institute of Ecology and Evolution). The R program (R Core Team 2023) **Acoustic Target Data Cleanup.R** was used to compile and clean the import data, which were unchanged by the process. R version 4.3.1 was used, and RStudio 2024.09.1 Build, 394.

Note that the DFO staff who assisted with the data extraction and cleaning (i.e., Howard Stiff and Athena Ogden) were not part of DFO's Acoustic and Trawl Survey (ATS) field crews that collected the data; therefore, when there was unclarity in some of the input data files (e.g., duplication of data records), some decisions about how to clean the data were based on educated guesses. These decisions are documented in the accompanying document, "**DFO-LEP Acoustic Target Data Issues (1977-2007).pdf**".

Cleaned Data Products

The cleaned acoustic survey data records (i.e., target counts by ats_year, lake, date, survey_type, transect and depth) can be found in CSV file: **Target_OUTPUT_data_FINAL_(1977_2007).csv**.

An inventory of the cleaned acoustic surveys (summarized by ats_year, lake, date, survey_type) can be found in CSV file: **Target_OUTPUT_data_INVENTORY_(1977_2007).csv**.

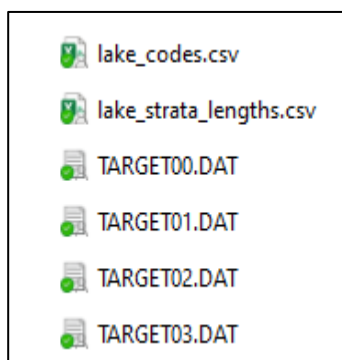
R Program Execution

For R code users who are interested in re-executing the **Acoustic Target Data Cleanup.R** program (for the purposes of, for example, regenerating the cleaned output after revising the R code to ensure inclusion of an omitted survey), the sub-directory structure containing the input files, program files and the output files should first be set up to include the following folders (where "..." represents the user's RStudio project directory):

1. ... \ACOUSTIC_TARGETS\code (containing "Acoustic Target Data Cleanup.R")
2. ... \ACOUSTIC_TARGETS\data (containing TARGETyy.DAT input data)
3. ... \ACOUSTIC_TARGETS\output (containing output CSV files, including the cleaned data)
4. ... \ACOUSTIC_TARGETS\output\figures (containing a facet plot of the total number of acoustic surveys by lake and year in the cleaned data)
5. ... \ACOUSTIC_TARGETS\output\archive (reserved for test runs of the R script)

Input Data

The following screen capture shows some of the input files contained in the \data folder.



The CSV files (lake_codes.csv and lake_strata_lengths.csv) are lookup tables used by the R program, with lake-specific codes and dimensions that are included in the cleaned output data.

The rest of the files are raw TARGETyy.DAT files. They are sequential files from "TARGET77.DAT" to "TARGET07.DAT".

Not all TARGETyy.DAT files are shown in this image.

Acoustic Target Data Cleanup.R saves nine different CSV files into the **...\ACOUSTIC_TARGETS\output** folder (right), including:

The compiled “uncleaned” data plus inventory:

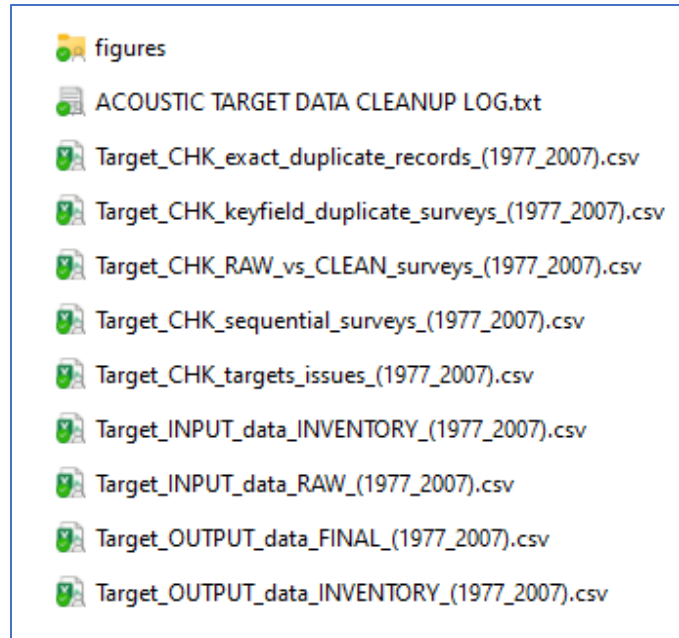
- Target_INPUT_data_RAW
- Target_INPUT_data_INVENTORY

The “cleaned” output data:

- Target_OUTPUT_data_Final and
- Target_OUTPUT_data_Inventory

And various intermediate “check” files, containing lists of data issues encountered or resolved during program execution:

- Target_CHK_exact_duplicate_records – list of records removed due to duplication of data across all fields or columns
- Target_CHK_keyfield_duplicate_surveys – list of surveys duplicated on key fields (lake, survey_date, survey_type, sounder_type)
- Target_CHK_sequential_surveys – list of surveys that occur within one day of each other for the same lake, representing possible replicate surveys (change in field survey method (e.g., sounder type), or lab analysis method (e.g., software used, technician performed)).
- Target_CHK_targets_issues – list of remaining records (if any) with anomalous data issues
- Target_CHK_RAW_vs_CLEAN_surveys – inventory of records organized by whether they exist only in TARGETty.DAT files (i.e., omitted from cleaned dataset), only in cleaned data (programmatic additions or key-field revisions), or in both.



The contents of these files are further described in “DFO-LEP Acoustic Target Data Issues (1977-2007).pdf”.

Note that the **ACOUSTIC TARGET DATA CLEANUP LOG.txt** file shown above contains the RStudio console output of a successful run of the R program (not automatically saved to the \output folder).

Output Data Review

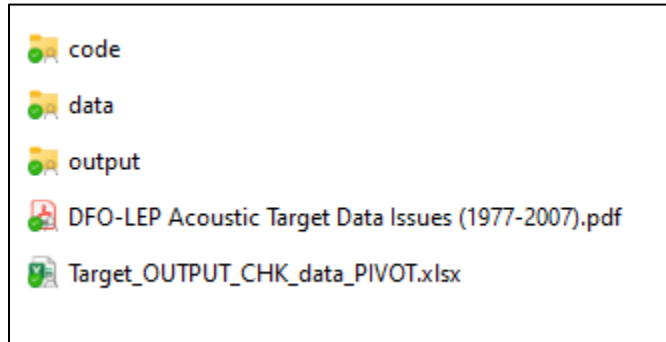
While the CSV files can each be opened in MS-Excel or read into R for review and further analysis, a quick look at the R program’s output files can be obtained with the included Excel workbook:

“Target_OUTPUT_CHK_data_PIVOT.xlsx”. This workbook contains multiple worksheets, each automatically linked to a corresponding CSV file for ease of data display, filtering, and pivot summary.

Once initial setup is performed (see details in section “[To Use the Target Output Checking Data Workbook](#)” in “DFO-LEP Acoustic Target Data Issues (1977-2007).pdf”), the data in “Target_OUTPUT_CHK_data_PIVOT.xlsx” worksheets are automatically updated with the contents of the CSVs upon opening the workbook or refreshing the data.

Additional Metadata Included as a ZIP File

The zip file, **DFO-LEP Acoustic Target Data (1977-2007).zip**, contains the following folders, following the subdirectory structure of the RStudio project employed to run the Acoustic Target Data Cleanup.R program:



Descriptions of Variables

ats_year: the value of this variable is the egg emergence year of the dominant cohort of juvenile Sockeye salmon (i.e., age-0), but the “ats_year” continues from April 1 in year X to March 31 in year X+1. Before April 1 in year X+1 is when the majority of the age-0 juveniles leave the lake as smolts. Note that juvenile Sockeye in different lakes might emerge (and recruit to the sampling gear) at different times of year, and leave the lake as smolts at different times of year. Therefore, the March 31/April 1 cut-off may not apply to all lakes; the analyst should review published data, technical, and manuscript reports in the Canadian Reports of Fisheries and Aquatic Sciences series (available at DFO libraries) to determine these dates.

lake_code: Lake Enrichment Program code of the lake where data collection occurred

lake: Lake Enrichment Program name of the lake where data collection occurred; lake name retrieved from “lake_strata_lengths.csv”

survey_date: date when the survey was performed, converted from a text format (YYMMDD) in the input files to ISO date format (YYYY-MM-DD)

target_survey_code: - numeric code classification taking the values 1 for JUVENILE or 2 for ADULT survey types, assigned in code. See target_survey_type.

target_survey_type: This survey type indicates the primary/intended type of the survey, ADULT or JUVENILE. This classification is derived from detecting phrasing in either the acoustic_survey_notes or acoustic_survey_comments fields indicating “ADULT COUNT”, “ADULT SURVEY”, “ADULT ESTIMATE”, or similar text. In the absence of specific text indicating an ADULT survey type, the default classification was JUVENILE. See target_survey_code.

survey_year: calendar year that the survey was performed

survey_month: month when the survey was performed

transect: specific transect number associated with a lake survey location

depth_code: numeric integer, one for every unique depth stratum (i.e., if depth_stratum = “2-5” then depth_code = 1, “3-5” is 2, etc.), to classify the lake depth associated with a target count

depth_min_m: minimum depth range (in meters) associated with the depth_code at which targets were ensonified

depth_max_m: maximum depth range (in meters) associated with the depth_code at which targets were ensonified

targets: number of “fish-like” acoustic targets detected (ensonified) in a specific depth range along a particular lake transect

prop_sockeye: proportion of targets that were determined to be Sockeye targets; NA means that the proportion Sockeye was missing in the input data. In addition, if the proportions of Sockeye and Stickleback were both zero in the input data, both proportions were set to NA in code. When the given proportions did not add up to 1.0 +/- 0.01, we have flagged this in the data_issues column. Note: we do not actually know where proportions reported in the TARGET files came from, or how they were derived. We do not place much stock in them, but rely on trawl sampling data for species composition.

prop_stickleback: proportion of targets that were determined to be Stickleback targets; NA means that the proportion Stickleback was missing in the input data. In addition, if the proportions of Sockeye and Stickleback were both zero in the input data, both proportions were set to NA in code.

prop_total: sum of the given proportions of Sockeye and Sticklebacks

transect_length_m: length of transect in units of meters

area_ha: lake area in hectares at a given depth level, which is dependent on the bathymetry of the lake

sounder_code: numeric code of the echo-sounder that was used to collect the data (1 = FURUNO or 2 = SIMRAD; BIOSONICS uncoded)

sounder_type: text field of type of sounder used to collect the data (FURUNO, SIMRAD, BIOSONICS)

sounder_gain: sensitivity of the sounder

sounder_issues: output comment field associated with either sounder code versus sounder type discrepancies, or sounder gain value missing or erroneous

acoustic_survey_notes: metadata provided in the input file regarding the acoustic survey

acoustic_survey_comments: as given in the input data, these are comments that usually include the type of survey that was done or by whom. There are a few comments that are truncated; these are truncated in the input file.

data_issues: comment containing one or more issue(s) that were found in the data during data compilation by R program

key_field_replicate: see section Handling Duplicate and Replicate Surveys, below for explanation

sequential_date_replicate: see section Handling Duplicate and Replicate Surveys, below for explanation

acoustic_source_file: source file name (TARGETyy.DAT, where yy = two-digit value representing individual years from 1977 to 2007) from which the data were imported/compiled.

line_number: line number of a survey or record in the source file.

Handling Duplicate and Replicate Surveys

In general, we define **duplicates** as identical records on all fields, whereas **replicates** are records that duplicate some fields but not others. Further details below.

We define **duplicate records** as records with identical values across all meta-data and data fields. We define **exact duplicate surveys** as surveys in which all records are duplicates. Duplicate surveys are identified in code and omitted from the final output of cleaned data, as described below.

A survey may be defined as a **key-field replicate** if another survey exists for the same key fields, i.e., *lake*, *survey date*, *survey type* (e.g., JUVENILE vs ADULT) and *sounder type* (e.g., FURUNO vs SIMRAD), but differs in other ways: e.g., lab technique (processor, methodology, software settings), or exhibits what are assumed to be updates to the data.

A survey may be defined as a **sequential-date replicate** if another survey exists for most of the same key fields, i.e., *lake*, *survey type* (e.g., JUVENILE vs ADULT) and *sounder type* (e.g., FURUNO vs SIMRAD), but differs by having a survey date that sequentially follows another survey. This method of saving two survey datasets that are nearly exactly similar may have been a simple 'desktop protocol' to differentiate data that differed in other ways: e.g., lab technique (processor, methodology, or software settings), or updates to the data. However, this has not been clearly documented. Rarely would re-surveying a lake on sequential days be executed in the field for the same purpose (i.e., stated survey type) with the same equipment (sounder) -- although Hobiton Lake and Kennedy Lake may be exceptions where experimental re-surveying occurred.

All surveys exhibiting key-field or sequential-date replication have been visually examined (see DFO-LEP Acoustic Target Data Issues (1977-2007).pdf) and, where possible, as based on survey comments/notes, the preferred survey dataset has been identified, and others marked for omission, from the final output data in the R code (see section of R code beginning with **# Clean up and flag data issues in data_issues field.**)

Exact Duplicates - If two duplicate surveys with identical data were found in the input data, one of them was deleted by the R code (using the `distinct()` function; see lines 1045 – 1066). A list of exact duplicates is generated at lines 1056-1060, and are exported to CSV in the output CSV file `Target_CHK_exact_duplicates_(1977-2007).csv` and can be easily reviewed in `Target_OUTPUT_CHK_data_PIVOT.xlsx`. Commenting out lines 1063-1065 will preserve the duplicate surveys in the output files, if desired.

Key-field Replicates – the **key_field_replicate** field has values of “NA” or “Replicate survey exists in the original data for this lake, date, sounder, transect and depth.” These replicates may be due to meta-data differences, e.g., different observers (i.e., echogram readers) generating different target values for a given transect and depth, or changes in species composition, or differences in target counts due to changes in echosounder software settings.

All records with apparent duplicates on these fields were examined, and if survey notes/comments revealed definitively which survey dataset was preferred, the other(s) were filtered from the data in the R code, as described below. Data removals via the R code can be revoked by commenting out appropriate portions of lines 436-504 in the R program. A list of remaining key-field replicate surveys can be found in the output CSV file `Target_CHK_key_field_replicates_(1977-2007).csv`, and can be easily reviewed in `Target_OUTPUT_CHK_data_PIVOT.xlsx`.

Sequential-Date Replicates – the **sequential_date_replicate** field has values of “NA” or “Replicate or potential duplicate (with sequential survey dates) exists in the original data.” These replicates may be due to different observers (i.e., echogram readers) generating different target values for a given transect and depth, or changes in species composition, or changes in software settings, but saved with a slightly different date to differentiate from the original survey dataset.

All surveys exhibiting apparent sequential-date replication were examined (see DFO-LEP Acoustic Target Data Issues (1977-2007).pdf), and if survey notes/comments revealed definitively which survey dataset was preferred, the other(s) were filtered from the data in the R code, as described in DFO-LEP Acoustic Target Data Issues (1977-2007).pdf. A list of remaining sequential-date replicates can be found in the output CSV file Target_CHK_sequential_date_replicates_(1977-2007).csv, and can be easily reviewed in Target_OUTPUT_CHK_data_PIVOT.xlsx. Again, data removals via the R code can be revoked by commenting out appropriate portions of lines 436-504 in the R program.

References

- Gjernes, T. W., K. D. Hyatt and D. P. Rankin. 1986. Development of specifications for the use of an uncalibrated echosounder to estimate populations of small limnetic fish. Can. Tech. Rep. Fish. Aquat. Sci. 1509: 35p.
- Hyatt, K. D., D. Rutherford, T. Gjernes, P. Rankin, and T. Cone. 1984. Lake Enrichment Program: Juvenile sockeye unit survey guidelines. Can. Ms Rep. Fish. Aquat. Sci. 1796: 84p.
- Hyatt, K.D., D.P. Rankin, and B. Hanslit. 2000. Acoustic and trawl based estimates of juvenile sockeye salmon (*Oncorhynchus nerka*) production from 1976 - 1999 brood year adults returning to Smith Inlet and Long Lake, British Columbia. PSARC Working Paper S00-21, DRAFT. 50p. (This draft document is available from the Salmon in Regional Ecosystem Program at Fisheries and Oceans Canada; contact Patrick.Thompson@dfo-mpo.gc.ca.)
- R Core Team. 2023. R: A Language and Environment for Statistical Computing. R Foundation for Statistical Computing, Vienna, Austria. <<https://www.R-project.org/>>

End of READ_ME Section

Introduction - Acoustic Target Data Issues

The purpose of this document is to outline the data issues encountered while processing the data and to document the various outputs provided by the R program, including (i) information on the raw data inputs and their processing, (ii) the handling/processing of certain data issues such as duplicates and potential replicates, and finally, (iii) the cleaned output data. Examples of duplicates and replicates as defined in the Read_Me section are described in detail, and other data issues associated with echosounder type/use are outlined as well.

While the data team is confident that the effort put into cleaning the Acoustic Targets data has yielded a high quality dataset for immediate use (i.e., **Target_OUTPUT_data_FINAL_(1977_2007).csv**), all users should familiarize themselves with the information below regarding the data issues encountered and the programmatic interventions and omissions implemented to achieve the final dataset.

As described above in the Data Cleaning section, all output CSV files can each be opened in MS-Excel or read into R for further review and analysis. However, the included Excel workbook:

“Target_OUTPUT_CHK_data_PIVOT.xlsx” can also be used to review the R program’s output files. This feature is described in the Post-Processing Data Review section below.

Post-Processing Data Review

To facilitate review of the Microsoft® Excel® workbook¹ is available containing worksheets linked to specific CSV files generated as output by the R program which is linked. See **Target_OUTPUT_CHK_data_PIVOT.xlsx** [Last updated: 02-Nov-25].

This Excel workbook should be stored in the user’s <LDP ATS Rescue> R project subdirectory. The workbook contains readily accessible copies of the saved CSV datasets generated by the **Acoustic Targets Cleanup.R** program. Each CSV file in the project Output directory is ‘hot-linked’ to a corresponding Excel workbook worksheet in the **Target_OUTPUT_CHK_data_PIVOT.xlsx** workbook. The worksheets are grouped (and worksheet tabs colour-coded) by three utility-types, specifically:

- **INPUT DATA** for *Target_INPUT*.csv* files – **black** tabs
- **DATA CHECKS** for *Target_CHK*.csv* files – **purple**, **green**, **blue**, and **gray** tabs
- **OUTPUT DATA** for *Target_OUTPUT*.csv* files – **yellow** tabs.

See the **INFO** tab in the **Target_OUTPUT_CHK_data_PIVOT.xlsx** for more information (Table 1 shows some of the columns in the “INFO” tab).

Worksheet names in **Target_OUTPUT_CHK_data_PIVOT.xlsx** prefixed with “**PIVOT**” are ‘child’ worksheets containing PIVOT-table-based filtered summary information of a ‘parent’ worksheet. See Table 1 (and/or the “INFO” tab) for a list of parent worksheets (and their corresponding CSV source files), and subsidiary PIVOT table child worksheets.

¹ Microsoft® Excel® for Microsoft 365 MSO (Version 2502 Build 16.0.18526.20546) for 64-bit computers.

Most worksheets also have built-in **SLICER** tables to facilitate visual filtering of the worksheet data (e.g., by lake, survey type and date, sounder type, etc.). The slicer tables can be revised or added to by the user.

To Use the Target Output Checking Data Workbook

After each successful execution of the **Acoustic Target Data Cleanup.R** program^{2,3}, open the **Target_OUTPUT_CHK_data_PIVOT.xlsx**, click the **Data** menu in the ribbon and select **Refresh All** to update all Target* parent worksheet 'hot-links' and their child PIVOT table worksheets⁴.

Review the PARENT tables to explore the individual CSV output tables (Table 1, below). Use existing (or create new) **SLICER** filters to drill down to records for specific lakes and surveys, or use associated **PIVOT** worksheets (where supplied, or create your own) to filter and review summary records.

- The **Target_INPUT_data_RAW** and **Target_INPUT_data_INVENTORY** worksheets display pre-processed input data imported by the R program from the raw data in the annual TARGETyy.DAT files (where yy = year).⁵
 - **Target_INPUT_data_RAW** displays the imported target data records and species proportions (Sockeye and Stickleback only) found in **Target_INPUT_data_RAW (1977_2007).csv**, by source file, lake code, lake comment, survey date, depth range (m), and transect.
 - **Target_INPUT_data_INVENTORY** displays an inventory of the imported target data by source file, lake code, lake comment, and survey date, found in **Target_INPUT_data_INVENTORY_(1977-2007).csv**.
- The **Target_CHK_*** worksheets document the duplicates analysis.
 - **Target_CHK_exact_duplicate** tab lists records identified as exact duplicates on all fields and thus omitted from further data-checking and output.
 - The **Target_CHK_Key_Field** and **Target_CHK_Sequential** tabs list any *remaining* potential duplicate/replicate survey data records that were identified by the code – as described below in sections *Key-Field Replicate Investigations* and *Sequential Survey Dates – Potential Duplicates*.

Note that a first pass through the data by the program was used to identify *initial* potential duplicate/replicate records, and where deemed appropriate, was used to include R code to omit likely duplicates or obsolete replicates. These omissions may be

² Successful execution of program displays execution time in the console window in RStudio, such as:
Acoustic Target Data Cleanup program complete.
Execution time: 5.88 minutes [01-Nov-2025]

³ See **Acoustic Target Data Cleanup Log.txt** in the **output** sub-directory for a typical log file for a successful execution of the R program.

⁴ Problems with linking worksheets to the CSVs? This is a likely event upon initial setup, and will also occur any time the location of the Output directory changes, but should not occur upon subsequent **Acoustic Target Data Cleanup.R** program runs. To refresh links, click the **Data** menu in the ribbon, and select **Queries and Connections** menu item. This will bring up the Queries window - right-click on any Query in the window, and choose **Edit** to bring up the **Power Query Edit** window. Select **Data Source Settings** in the menu to list all data sources. Select each data source and click the **Change Source Settings** button, then browse to location of corresponding CSV.

⁵ Note that *.DAT files can also be viewed using a text editor.

revoked in the R code by commenting out the associated filter statements in the section beginning with the comment (approx. line 437 - 504):

```
# Filter out duplicate survey data identified in LDP Data Issues.PDF
```

- The **Target_CHK_RAW_vs_CLEAN_surveys** worksheet provides an inventory of the lake/survey-date combinations in the analysis, classified by **source** as to whether the survey was:
 - (A) found only in the raw data (i.e., omitted entirely from outputs, or modified in the code to alter one or more key fields);
 - (B) found only in the final cleaned data (i.e., a function of programmatic revisions to the key-fields for lake and survey); or
 - (C) found in both raw import data and cleaned output data inventories.
- The **Target_OUTPUT_data_*** worksheets display the final “cleaned” acoustic target data, corresponding to analogous **Target_OUTPUT_data_*_(1977-2007).csv** files outputted by the R program in the Output subdirectory.
 - **Target_OUTPUT_data_INVENTORY** tab displays an inventory of the imported target data by ATS year, lake code, survey type (juvenile vs. adult), sounder type, and survey date. Each record’s original raw data source file and various acoustic survey comment fields and notes about potential replicates are included.
 - **Target_OUTPUT_data_FINAL** tab displays the cleaned target data records and species proportions (Sockeye and Stickleback only) by ATS year, lake code, lake comment, survey date, depth code, and transect. The associated transect length (m) and area (m²) at depth (m) are included. Sounder type and other acoustic survey notes derived from the input files are included. Specific issues associated with a given record are listed in the data_issues field. Potential duplicate/replicate data issues are noted in separate fields under column headers key_field_replicate and sequential_date_replicate. The TARGETyy.DAT source file name and the line number in the source file are included in case the user needs to return to the source data for review or verification purposes.

GROUPING	PARENT WORKSHEET	SOURCE CSV FILE	CHILD WORKSHEET
INPUT DATA	Target_INPUT_data_RAW	Target_INPUT_data_RAW_(1977-2007).csv Compilation of imported data records from TARGET*.DAT files.	
	Target_INPUT_data_INVENTORY	Target_INPUT_data_INVENTORY_(1977-2007).csv Inventory listing of unique surveys imported from TARGET*.DAT files.	
DATA CHECKS	Target_CHK_exact_duplicates	Target_CHK_exact_duplicates_(1977-2007).csv List of exact duplicate records (i.e., duplicates across all fields); removed from data.	Pivot Exact Dups Removed
	Target_CHK_Key_Field_Replicates	Target_CHK_Key_Field_Replicates_(1977-2007).csv List of duplicates on lake, survey date, transect and depth. Matches were visually compared (see LDP Data Issues.docx) and if survey notes or comments were sufficient to distinguish the most representative survey, code was added to ACOUSTIC TARGET DATA CLEANUP.R to omit the other survey data. Remaining key-field replicates are listed in this Excel table.	Pivot Key_Field Replicate List Pivot Key_Field Replicates
	Target_CHK_sequential_surveys	Target_CHK_sequential_surveys_(1977-2007).csv Data associated with replicate surveys that differ by one sequential date, ignoring replicate field survey using different sounders, or for different purposes (e.g., JUVENILE vs ADULT surveys). Beyond that, data managers sometimes replicated surveys (applying a different date) to compare different observer results, or to apply different species proportions (using the same target counts) according to survey notes. Matches were visually compared (see LDP Data Issues.docx) and if survey notes or comments were sufficient to distinguish the most representative survey, code was added to ACOUSTIC TARGET DATA CLEANUP.R to omit the other survey data. Remaining key-field replicates are listed in this Excel table.	Pivot Seq Date Replicates
	Target_CHK_RAW_vs_CLEAN_surveys	Target_CHK_RAW_vs_CLEAN_surveys_(1977-2007).csv Compilation of inventory records, organized by whether the records were found only in the input TARGET*.DAT files (i.e., discarded as duplicates, etc), found only in the final cleaned up data (fixes and additions), or both (input data that have been cleaned for final output).	
OUTPUT DATA	Target_OUTPUT_data_FINAL	Target_OUTPUT_data_FINAL_(1977-2007).csv Compilation of QA/QC'd (cleaned) data records to be saved in Open Data forum.	
	Target_OUTPUT_data_INVENTORY	Target_OUTPUT_data_INVENTORY_(1977-2007).csv Listing of unique QA/QC'd surveys to be saved in Open Data forum.	

Table 1. Linked worksheets in Target_OUTPUT_CHK_data_PIVOT.xlsx and corresponding CSV output files from Acoustic Target Data Cleanup.R. See **INFO** tab in Target_OUTPUT_CHK_data_PIVOT.xlsx for more information.

Duplicates and Replicates Handling

For definitions of “duplicate” vs. “replicate”, see also the Read_Me section, above. ; however, specifically for handling of duplicates and replicates, see also the Post-Processing Data Review.

Note again that a first pass through the data by the program was used to identify initial potential duplicate/replicate records, and where deemed appropriate, was used to include R code to omit likely duplicates or obsolete replicates. Except for Exact Duplicates (records that are duplicates in all columns), data removals may be revoked in the R code by a user with some coding skills, by commenting out the associated filter statements in the section beginning with the comment (approx. lines 437 - 504).

Exact Duplicates

Duplicate records across all data fields.

Duplicate Surveys										
			TARGET83.DAT	TARGET86.DAT	TARGET90.DAT	TARGET95.DAT	TARGET99.DAT			
			1983	1986	1990	1995	1999			
			FURUNO FM 22		SIMRAD					
		NA	PARTIAL OVERCAST FULL MOON TRANSECTS 1-5	100%, gusty, chop, no moon, At: 16/17 0c (2145/040 Wt: 19 0c, JC 861020, targ overlap, poor spp break-down, Juvenile.	60- 30%clld,lightwnd,dr izzle/mist,flmn,AT 7ST18.5degC	clr,brze,nwmn,AT1 6ST15.6degC. target banding around 20m.	difference between total fish and sx-stk are sclpns.	Clear, Calm, Surface Smooth, AT 5.9, ST 10.8, No M 18Sep95/07Nov95/ AR/EYM/20111/1/0/ 0/11/86/0/	ovrcst,nownd,mod rn,nomn,AT12ST14 degc. 23AUG95/11SEP95 /JH/EYM/20128//1/ 0/0/7/96/405/.	nomn,lgtsnw,ocst, brgtaurora17th 140c199/03Nov99/ PR/EYM/20128/1/0/ 0/18/27/0.
Henderson Lk	Juvenile Survey processed by PR, 07May91.			30-Aug						
	processed by PR, 08May91.				05-Oct					
Long Lk (A)	NIGHT SURVEY processed by JH, 11Sep95.	18-Jul	24-Aug		20-Jun					
Owikeno Lk (D)	processed by SM, 14May91.								23-Aug	
Skidegate Lk ne	NA						19-Aug			
Tatsamenie Lk	processed by PR,03Nov99.							18-Sep		
										14-Oct

Screenshot listing all exact duplicates found. One instance of each duplicate record set was removed during clean up of the target data, but retained in the source files.

Key-Field Replicate Investigations

Surveys that are not exact duplicates, but have the same key field meta-data (i.e., lake, survey_date, sounder, processor).

These potential duplicates were compared in spreadsheet pivot tables, and handled (or ignored) in the program as described below for each case, below.

See [Target_OUTPUT_CHK_data_PIVOT.xlsx](#).

Key-Field Replicate List

lake	ats_year	survey_date	source_file
Hobiton Lk	1989	1989-09-19	TARGET89.DAT
		1990-02-22	TARGET89.DAT TARGET90.DAT
Johanson Lk	1994	1994-08-25	TARGET93.DAT
			TARGET94.DAT
Kennedy Lk (Clayoquot Arm)	1984	1985-01-23	TARGET84.DAT
			TARGET85.DAT
Owikeno Lk (B)	2003	2004-02-04	TARGET03.DAT
			TARGET06.DAT
Sustut Lk	1994	1994-08-23	TARGET93.DAT
TARGET94.DAT			
Tahltan Lk		1994-09-18	TARGET93.DAT
			TARGET94.DAT
Tatsamenie Lk	1994-09-13	TARGET93.DAT	
		TARGET94.DAT	
	1997	1997-06-26	TARGET96.DAT
			TARGET97.DAT
Trapper Lk	1994	1994-09-11	TARGET93.DAT
			TARGET94.DAT
Tuya Lk		1994-09-02	TARGET93.DAT
			TARGET94.DAT

Hobiton Lake - lake_code 6

lake	Hobiton Lk			
ats_year	All			
Sum of targets	Survey			
	Replicate exists for this lake, date, transect and depth			
	1989-09-19		1990-02-22	
	TARGET89.DAT		TARGET90.DAT	
	SIMRAD			
	JUVENILE			
	Processed by JC	processed on 08Nov89 by RT.	processed JF, 14Jan91	
Transect	clr,5/8mn,at11.5,st18C	clr,SEwnd,1/2mn@2145,AT11.5ST18@2220.	processed by JF, 14Jan91.	processed by JF, 14Jan91.
1	81	74	27	27
2	53	46	42	42
3	80	77	115	115
4	599	545	275	275
5	136	119	93	93
6	172	160	39	39
7	253	232	93	93
8	170	149	364	364
9	209	191	145	145
10	647	668	356	356
11	751	713	415	415
Grand Total	3151	2974	1964	1964

- 89-09-19 - This appears to be a true replicate count of the same echosounder data by different processors (JC and RT), as shown in row 10. No removals advised; both record sets retained in the output data.
- 90-02-22 – version in TARGET90.DAT appears to be a duplicate (placed in wrong TARGET.DAT file) – **deleted**

Johanson Lake - lake_code 226

lake	Johanson Lk	
ats_year	All	
Sum of targets	Survey	
	Replicate exists for this lake, date, transect and	
	1994-08-25	
	TARGET93.DAT	TARGET94.DAT
	SIMRAD	
	JUVENILE	
	Processed by A.R. Dec14/94	
Transect	CC 0%, PPT Nill, Wind Calm, Waves Calm, AT 4.9, ST EY-M 200069, DAT A83289	CC 0%, PPT Nill, Wind Calm, Waves Calm, AT 4.9, ST EY-M 200069, DAT A83289
1	4	4
2	52	52
3	62	62
4	68	68
5	30	30
6	2	2
Grand Total	218	218

- 94-08-25 – version in TARGET93.DAT appears to be a duplicate (placed in wrong TARGETyy.DAT file) – **deleted**

Kennedy Lk (Clayoquot Arm) - lake_code 40

lake	Kennedy Lk (Clayoquot Arm)		
ats_year	All		
Sum of targets	Survey		
	Replicate exists for this lake, date, transect and depth		
	1985-01-23		
	TARGET84.DAT		TARGET85.DAT
	FURUNO FM 22	SIMRAD	
	JUVENILE		
	JUVENILE SURVEY		
Transect	CALM, SURFACE & AIR TEMP. = 6 DEG. C. COUNTS BELOW 20 M, AT TRANS #2, ARE MINIMUM COUNTS, counted 850205 PR	CALM, WATER & AIR TEMP. = 6 DEG. C. COUNTED FEB. 6/85 BY PR. MINIMUM COUNTS: T6,20-30M; T3,40-50M; T4&7,60-70M	0/100%, calm, no moon, At: 5/9.5 0c (2045/1830) Wt: 6 0c, JC 851204.
1	1018	1307	2939
2	3957	2092	9898
3	1888	2330	4039
4	680	981	1708
5	1096	1774	2833
6	630	1300	1794
7	287	577	558
8	0	0	0
9	0	0	0
10	0	0	0
11	0	0	0
12	0	0	0
13	0	0	0
14	0	0	0
15	0	0	0
16	0	0	0
Grand Total	9556	10361	23769

- 85-01-23 - it appears that two field surveys were done for 85/01/23 and stored in TARGET84.DAT, likely valid intentional replicates for hardware comparison: one with Furuno, one with Simrad sounder.
- However, the third 'replicate' found in TARGET85.DAT bears little resemblance to either – this may be an incorrectly coded date or lake (***might possibly be Kennedy-Main (KMA) instead of Kennedy-Clay (KCA)***). It appears to have been processed much later in the year (December) by JC, whereas the TARGET84.DAT datasets were processed by PR (Rankin). No additional info from the meta-data notes were available to inform date assignment. Consult Survey Trip Reports (STRs) or SRE documents (if available) for further clues (these are documents that may be available in the electronic and/or hard copy files of the late Dr. Kim Hyatt; contact Patrick.Thompson@dfo-mpo.gc.ca at the Salmon in Regional Ecosystems Program at DFO). Not removed.

Owikeno Lake (B) - lake_code 229

lake	Owikeno Lk (B)	
ats_year	All	
Sum of targets	Survey	
	Replicate exists for this lake, date, transect and	
	2004-02-04	
	TARGET03.DAT	TARGET06.DAT
	SIMRAD	
	JUVENILE	
	Hanslit-Bachen Survey	
Transect	4Feb04/23Mar07/EYM/20069/0/0/0/20/937/217	4Feb04/23Mar07/EYM/20069/0/0/0/20/937/217
1	6370	6370
2	1751	1751
3	1524	1524
4	1284	1284
5	3916	3916
6	1469	1469
7	435	435
8	270	270
Grand Total	17019	17019

2004-02-04 – version in TARGET06.DAT appears to be a complete duplicate, placed in the wrong TARGETty.DAT file. Deleted.

Sustut Lk - lake_code 225

lake	Sustut Lk	
ats_year	All	
Sum of targets	Survey	
	Replicate exists for this lake, date, transect and	
	1994-08-23	
	TARGET93.DAT	TARGET94.DAT
	SIMRAD	
	JUVENILE	
	Processed by A.R. Dec.13/94	
Transect	CC 95%,Fog,Wind light,AT8,ST14,Moon Rise 22:39 EY-M 20069	CC 95%,Fog,Wind light,AT8,ST14,Moon Rise 22:39 EY-M 20069 Processed by A.R. Dec.13/94
1	48	48
2	132	132
3	501	501
4	243	243
5	19	19
Grand Total	943	943

94-08-23 - This appears to be a true duplicate (including *targets*, *spp. proportions*), erroneously included in TARGET93.DAT. TARGET94.DAT version has added info in the acoustic_survey_comment line (row 11). TARGET93.DAT version deleted.

Tahltan Lk - lake_code 64

lake	Tahltan Lk	
ats_year	All	
Sum of targets	Survey	
	Replicate exists for this lake, date, transect and	
	1994-09-18	
	TARGET93.DAT	TARGET94.DAT
	SIMRAD	
	JUVENILE	
	Processed by A.R. Dec15/94	
Transect	CC 0%, Ambient Light High A.T. 5.0, S.T. 12.1, Moo EY-M 20069, DAT 83289	CC 0%, Ambient Light High A.T. 5.0, S.T. 12.1, Moo EY-M 20069, DAT 83289
1	47	47
2	87	87
3	166	166
4	88	88
5	54	54
6	31	31
7	123	123
8	269	269
9	158	158
10	30	30
11	49	49
12	56	56
13	146	146
14	126	126
Grand Total	1430	1430

- 94-09-18 - This appears to be a complete duplicate (including *targets*, *spp. proportions*), erroneously included in TARGET93.DAT. **TARGET93.DAT version deleted.**

Tatsamenie Lk - lake_code 66

lake	Tatsamenie Lk			
ats_year	All			
Sum of targets	Survey			
	Replicate exists for this lake, date, transect and depth			
	1994-09-13		1997-06-26	
	TARGET93.DAT	TARGET94.DAT	TARGET96.DAT	TARGET97.DAT
	SIMRAD			
	JUVENILE			
	Processed by A.R. Dec 15/94		NA	
Transect	CC 100%, PPT Rain, Wind Mod, EY-M 20069 DAT 83289	CC 100%, PPT Rain, Wind Mod, EY-M 20069 DAT 83289	26Jun97/4Jul97/BH/EYM/1 0005/0/0/0/12/0/0/	26Jun97/4Jul97/BH/EYM/1 0005/0/0/0/12/0/0/
1	323	323	99	99
2	145	145	145	145
3	199	199	151	151
4	151	151	71	71
5	275	275	68	68
6	236	236	188	188
7	82	82	117	117
Grand Total	1411	1411	839	839

- 94-09-13 - This appears to be a complete duplicate, erroneously included in TARGET93.DAT. **TARGET93.DAT version deleted.**
- 97-06-26 - This appears to be a complete duplicate, erroneously included in TARGET96.DAT. **TARGET96.DAT version deleted.**

Trapper Lake - lake_code 67

lake	Trapper Lk		
ats_year	All		
Sum of targets	Survey		
	Replicate exists for this lake, date, transect and depth		
	1994-09-11		
	TARGET93.DAT	TARGET94.DAT	
	SIMRAD		
	JUVENILE		
	NA		
Transect	CC 100%, PPT Rain, Wind 30 kts, Wavec 50 cm, AT 8. EY-M 20069, DAT 83289	CC 100%, PPT Rain, Wind 30 kts, Wavec 50 cm, AT 8. EY-M 20069, DAT 83289	NA
1	18	18	18
2	14	14	14
3	9	9	9
4	90	90	90
5	44	44	44
Grand Total	175	175	175

- 94-09-11 – These appear to be complete duplicates, except for the survey notes. TARGET93.DAT version and the TARGET94.DAT version at line number 4541 deleted

Tuya Lake - lake_code 175

lake	Tuya Lk		
ats_year	All		
Sum of targets	Survey		
	Replicate exists for this lake, date, transect and		
	1994-09-02		
	TARGET93.DAT	TARGET94.DAT	
	SIMRAD		
	JUVENILE		
	Processed by A.R. Dec 94		
Transect	CC 100%, PPT Drizzle, Wind S.E. 30kts, AT 6.5, ST EY-M 20069, DAT 83289	CC 100%, PPT Drizzle, Wind S.E. 30kts, AT 6.5, ST EY-M 20069, DAT 83289	
1	368	368	
2	190	190	
3	62	62	
4	105	105	
5	182	182	
6	1	1	
7	669	669	
8	38	38	
9	116	116	
10	728	728	
Grand Total	2459	2459	

- 94-09-02 – These appear to be complete duplicates. TARGET93.DAT version deleted.

Sequential Survey Dates – Potential Duplicates

See the definitions of duplicates and replicates in the Read_Me section, above.

Includes surveys that have some of the same key field meta-data (lake, sounder, processor), but are on back-to-back survey_dates.

Many of these appear to be simple duplications of the data across the overnight span of the survey effort, since juvenile surveys usually occur in the evening and continue into the early morning hours the next day. It seems that in some cases, this resulted in a copy associated with the date of the survey start time and an additional copy for the next day when the survey was finished.

Other sequential survey_date issues appear to be erroneous inclusions of a single survey record set in two different TARGETyy.DAT files, possibly due to some confusion regarding the ATS-year to which to attribute a survey. For example, a survey from March 1991 should normally be included in TARGET90.DAT as it falls into the April'90 to March'91 ATS-year, but may show up erroneously in TARGET91.DAT. It is not clear why the duplicate was sometimes saved with a different date of one day, however.

In other cases, the sequential date occurrences appear to be bona fide replicate field surveys, testing different sounder types, or targeting juveniles versus adults (which probably involves revised sounder gain settings for larger targets). Usually ADULT surveys are noted in the TARGET*.DAT files in the acoustic_survey_comment field or survey_notes field, and, if the word ADULT SURVEY is found in these fields, the program applies a survey_type classification to the record, set to ADULT (default JUVENILE). Another clue is that Adult target counts are significantly fewer in number than Juvenile target counts around the same date.

Finally, there may be replicate echogram reading efforts 'back in the lab', by different processors (echogram readers), with different software (e.g., EYM versus EY500?), or with different goals (e.g., minimum versus full counts), yielding different target counts on or around the same date. It seems like it was common practice to save these replicates with a slightly different date.

Certain 'replicate' surveys involve revisions to the species proportions, based on updated trawl data, or adjustments for size bias in trawl observations. This might suggest that the uncorrected dataset should be omitted/deleted. The rationale for the extra dataset is not always made clear in the meta-data, however.

All sequential potential duplicates were exported to CSV and compared in spreadsheet pivot tables, and handled (or ignored) in the program as described below each case. Only obvious duplicate record sets are removed from the output dataset, via filtering mechanisms in the R code. Suggested removals are (usually) **marked in red** below in the screengrab and the explanatory text; where indicated, these have in fact been deleted from the final cleaned file, **Target_OUTPUT_data_FINAL_(1977_2007).csv**, that was provided to GC Open Data.

See the **Target_CHK_sequential_surveys** tab, or its source CSV file:
Target_CHK_sequential_surveys_(1977_2007).csv.

Brannen Lake

	A	B	C	D	E
1	lake	Brannen Lk			
2	ats_year	(All)			
3					
4	Tot Targets	Survey			
5		Replicate or potential			
6		1986-04-15 1986-04-16	survey_date		
7		JUVENILE	survey_type		
8		TARGET86.DAT	source_file		
9		SIMRAD	sounder_type		
10		2	sounder_code		
11		Juvenile Survey	acoustic_survey_comments		
			acoustic_survey_notes		
		20-100%, variable winds, showers, 1/4 moon, At: 5 0c (2100), Wt: 10 0c, JC 860416, Juvenile.	20-100%, variable winds, 1/4 moon, At: 10 0c (2100), Wt:10 0c, JC 860416, Changed spp compositon, Juvenile.		
12	Transect				
13	1	210	210	total targets by transect	
14	2	173	173		
15	3	221	221		
16	4	137	137		
17	5	280	280		
18	6	153	153		
19	Grand Total	1174	1174		

This appears to be a duplicate survey, but acoustic_survey_notes indicates that the 860416 version has revised species composition (prop_sockeye and prop_stickleback), which was verified in TARGET86.DAT. 86-04-15 version was deleted.

Cheewhat Lake

lake	Cheewhat Lk								
ats_year	(All)								
Tot Targets	Survey								
	Replicate or potential duplicate survey (sequ								
	1989-09-20	1989-09-21	1990-10-02	1990-10-03	1994-06-24	1994-06-25	1994-07-28	1994-07-29	
	JUVENILE								
	TARGET89.DAT		TARGET90.DAT		TARGET94.DAT				
	SIMRAD								
	2								
	processed by RT on Nov. 20'89.	processed by JC	processed by JF, 11Jan91.		Processed by S.T. July 7/94	processed by PR, 08Aug94.	EY-M#20069, RDAT# A83289	processed by PR, 08Aug94.	
clr,nownd,mnu p@23011/2full, at12.5,st18.5@ 2320 Sx were 84-0+,.05- 1+,.01-2+.	clear,1/2full,at 12.5,st18.5C@2 320 Sx were .85- 0+,.05-1+,.01- 2+	ovrcst,gstingwn d,hvyrn,AT11ST 18degC. sx 60% smll,40% lge. JF, 11Jan91.	ovrcst,gstingwn d,hvyrn,AT11ST 18degC. same as 901002. Trans#6 proportions set to 0 to improve var?	100%cc,wind- calm,no rain,AT16.2, ST18.5 degC Moon-up, sky bright but full overcast.EY- M#20111, Steve Toth 7Jul94.	Processed by PR, 08Aug94.	C.C. 100%, winds - calm, A.T. 20.2, S.T. 21.6 Recording level 6, surveyed by S.T.and A.R. Processed by ST, 03Aug94	Processed by PR, 08Aug94.		
Transect									
1	141	153	47	47	17	14	181	139	
2	169	171	131	131	16	14	296	213	
3	338	343	159	159	61	55	140	139	
4	168	158	50	50	201	177	53	57	
5	42	33	25	25	155	102	7	8	
6	649	677	446	446	208	186	592	470	
Grand Total	1507	1535	858	858	658	548	1269	1026	

lake	Cheewhat					
ats_year						
Tot Targets						
	ential dates)					
	1995-06-24	1995-06-25	1995-06-26	1995-07-28	1995-07-29	1995-07-30
	TARGET95.DAT					
	eym# 20069,20111. Dat#a61400. a28389	Eym#20069,2 0111, Dat#A61400, A23839	processed by PR, 19Jul95	NA	Processed by J.H. Aug11/95	NA
	clear,winds mod,waves 30cm,No moon, AT19.8,ST19.2 24Jun95/06Jul9 5/JH/EYM/2006 9/20111/1/0/1/ 4/246/0/	clear,winds mod,waves 30cm,No moon, AT19.8,ST19.2 24Jun95/07Jul9 5/ST/EYM/2006 9/20111/1/0/1/ 4/246/0/	24JUN95/19JUL 95/PR/EYM/200 69/20111/1/0/1 /4/246/0	28JUL95/23AU G95/ST/EYM/20 128//1/0/1/10/1 95/23	Air17.0,H2O21. 0,wind- none,waves- 0cm,rain- none,c.c. 28JUL95/11AU G95/JH/EYM/20 128//1/0/1/10/1 95/23	28JUL95/17AU G95/PR/EYM/20 128//1/0/1/10/1 95/23
Transect						
1	52	58	44	203	130	135
2	110	145	112	227	165	158
3	170	191	160	163	133	129
4	51	68	46	53	49	48
5	13	13	12	3	2	3
6	340	408	340	553	335	323
Grand Total	736	883	714	1202	814	796

These all appear to be bona fide replicates of some sort, via different lab processors, as based on the metadata provided in the originals.

None removed.

Devon Lake

lake	Devon Lk	▼
ats_year	(All)	▼
Tot Targets	Survey	▼
	Replicate or potential	
	1986-09-07	1986-09-08
	JUVENILE	
	SIMRAD	
	2	
	TARGET86.DAT	
	Juvenile Survey	
	0%. calm, no moon, At: 14.5/14 0c (2100/2235) Wt: 17 0c, JC 860909, targ overlap, no adults, Juvenile.	0%, calm, no ppt, no moon, At: 14.5/14 0c (2100/22 Wt: 17 0c, JC 860929, target overlap, no adults, Juvenile.
Transect		
1	822	99
2	783	295
3	477	175
4	1354	622
5	1986	968
6	1164	386
7	1529	335
Grand Total	8115	2880

Both survey analyses suggest some methodological change to the counting to deal with target overlap. No removals.

Eden Lake

lake	Eden Lk	▼
ats_year	(All)	▼
Tot Targets	Survey	▼
	Replicate or potential	
	1994-09-07	1994-09-08
	JUVENILE	
	SIMRAD	
	2	
	TARGET94.DAT	
	Ey-m#20111, DAT# A61400	
	C.C.10%, AT11. 8, H2O 16.5 Processed by S.T.091694	C.C.10%, AT11. 8, H2O 16.5 Processed by S.T.091694, species reassigned by pr, 22Feb95.
Transect		
1	111	111
2	118	118
3	241	241
4	198	198
Grand Total	668	668

- 94-09-07 and -08 datasets indicate change in species proportions by PR. 94-09-07 deleted.

Great Central Lake

lake	Great Central									
ats_year	(All)									
Tot Targets	Survey									
	Replicate or potential dup									
	1979-12-19	1979-12-20	1990-06-13	1990-06-14	1990-10-15	1990-10-16	1990-10-17	1990-10-18	1991-03-05	1991-03-06
	1991-03-07									
	FURUNO FM 22									
	1									
	TARGET79.DAT									
	TARGET90.DAT									
	NA		processed by PR on 05Dec90.	processed by JF, 15Jan91.	processed by JF 10Jan90.	processed by sm, 01Aug91.	processed on 03Dec90 by PR.		PROCESSED BY PR, 18MAR91.	
	NA	Ratios adjusted for size selectivity.	5%CLD,CALM, mup@01:00AT16ST16degC. EYM#20111. Processed by PR, 05Dec90.	processed by JF, 15Jan91.	ovrcst,lightwnd, shvshwrs,AT6ST3degC. 170ct90 survey, processed by JF 10Jan90.	NA	ovrcst,lightwnd, shvshwrs,AT6ST3degC. species breakdown suspect-insufficient trawling	ovrcst,lightwnd, shvshwrs,AT6ST3degC. species breakdown suspect-insufficient trawling assignment changed	75%CLD,CALM, FLLMN,SNWFL URRIES. BAD WEATHER AT START AFFECTED AMOUNT OF TRAWLING.	75%CLD,CALM, FLLMN,SNWFL URRIES. BAD WEATHER AT START AFFECTED AMOUNT OF TRAWLING.
Transect										
1	1581	1581	257	233	564	482	500	500	1124	1124
2	2111	2111	949	1190	1434	1410	1363	1363	1986	1986
3	1352	1352	457	560	1451	1364	1420	1420	1689	1689
4	796	796	613	754	1120	985	1076	1076	1505	1505
5	1221	1221	1169	1213	1644	1317	1398	1398	4696	4696
6	1096	1096	597	553	1253	1057	1096	1096	2889	2889
Grand Total	8157	8157	4042	4503	7466	6615	6853	6853	13889	13889

lake	GCL									
ats_year										
Tot Targets	SIMRAD									
	2									
	TARGET94.DAT		TARGET95.DAT		TARGET98.DAT		TARGET00.DAT		TARGET01.DAT	
	TARGET03.DAT		TARGET05.DAT		TARGET07.DAT		TARGET09.DAT		TARGET11.DAT	
	Processed by A.R. Jan 30th, 1995	processed by pr, 01Feb95.	NA	eym#10005, Dat#a61400	Rankin Hanslit Survey	Rankin - Hanslit Survey (HS testing)	Rankin - Hanslit Survey	Rankin-Hanslit Survey	Rankin-Hanslit Survey	Processed by PR, 15Feb2005.
	CC 100%, No Moon, Wind Calm to Light, Waves nil, PP EYM 20069, DAT A83289 Processed by A.R. Jan 30th, 1995	P. Rankin estimate, some target overlap.	17JUL95/28AUG95/PR/EYM/10005/1/0/1/11/4 98/12	c.c.0%,winds calm,waves0c m,air=19.7,h2o=23.3	2Dec98/5Jan98/BH/EYM/20069/0/0/0/11/120/2 8	2Dec98/5Jan98/BH/EYM/20069/0/0/0/11/120/28	30Nov00/8Jan01/BH/EY500/0/0/0/10/233/80 - duplicate target counts but spp proportions different (hs 25/10/07)	EY500 Used 4Dec01/22Jan02/BH/EY500/0/0/0/11/123/18. Adjusted for size selectivity.	EY500 Used 4Dec01/22Jan02/BH/EY500/0/0/0/11/123/18. Adjusted for size selectivity.	nomn;100pct cc;modEwmd; nomn;AT-ST3.9. 19Jan04/15Feb04/pr/EY500/0/0/0/14/18 1/168. Adjusted for size-bias trawl effect.
Transect										
1	1222	2814	132	129	777	777	878	878	372	969
2	2057	1809	945	991	2895	2895	1554	1554	641	2299
3	2212	1719	1561	2031	2191	2191	1621	1621	480	1359
4	3106	2286	1505	1865	4356	4356	3199	3199	936	2446
5	4318	3099	2653	3817	3723	3723	4385	4385	603	3825
6	4274	3283	1484	1998	3344	3344	2922	2922	379	3097
Grand Total	17189	15010	8280	10831	17286	17286	14559	14559	3411	13995

- 79-12-19 – deleted because 79-12-20 species proportions adjusted for size selectivity
- 90-06-13 & 90-06-14, 90-10-15 & 90-10-16: bona fide replicates; no deletions necessary
- 90-10-17 and -18 – species proportions adjusted as compared to 90-10-16; 90-10-17 deleted.
- 91-03-05 and 91-03-06 – deleted, as duplicates of 91-03-07 which was adjusted for size selectivity
- All 1995 data shown here – bona fide replicates, no deletions necessary
- 98-12-02 and 98-12-03 – duplicate target numbers, but species proportions differ, with no comment on which analysis is best. 98-12-02 deleted, assuming the fix was in 98-12-03.
- 2000-11-29 and 2000-11-30 - duplicate targets, but species proportions differ, with 2000-11-29 adjusted for size selectivity. 2000-11-30 deleted.
- 2001-12-04 – deleted; 2001-12-05 had spp. proportions adjusted for size selectivity.
- 2004-01-22 – deleted; 2004-01-23 had spp. proportions adjusted for size selectivity.

Henderson Lake

lake	Henderson L.										
ats_year	(All)										
Tot Targets		Survey									
		Replicate or potential duplicate survey									
		1986-09-23		1986-09-24		1993-02-25		1993-02-26		1993-07-16	
								1993-07-17		1994-06-26	
										1994-06-27	
										1995-06-20	
										1995-06-21	
										JUVENILE	
										SIMRAD	
										2	
		TARGET86.DAT		TARGET92.DAT		TARGET93.DAT		TARGET94.DAT		TARGET95.DAT	
		Juvenile Survey		Counted Mar.31/93 SM		NA		Processed by S.T. July 7/94		processed by PR, 20Jul94.	
										NA	
										15-20m strata zeroed, poor stylus contact.	
		60-100%, strong winds, 1/2 moon by 0000, At 12 oc (1957), Wt: 17 Oc, JC 861021, targ overlap poor, targ rendition poor, Juvenile.		60-100%, strong winds, 1/2 moon by 0000, At 12 oc (1957), Wt: 17 Oc, JC 861021,correct ed for overlap 18Jan89 PR.		clr, ctm, .1moon @1830-2100, at -3.0, wt +3.5 deg EY-M 2011, Log40, Dynatine ON.		clr, ctm, .1moon @1830-2100, at -3.0, wt +3.5 deg species assignment by strata changed		BASED ON COUNTS OF TRANSECTS 1-7 ONLY.	
										BASED ON COUNTS FOR 18 TRANSECTS.	
										100%cc,no rain,wind 5-10kts,AT14.2,S T14.3 EY-M/2011,2012 8. Processed by S.T. July 7/94	
										0-100%cid,light wmd,AT14.4-17.2WT14-15. some target overlap @head of lake processed by PR, 20Jul94.	
										21JUN95/06JU L95/JH/EYM/20 069/20111/1/0 /1/6/98/0	
										21Jun95/10Oct 95/pr/EYM/200 69/20111/1/0/ 1/6/98/0	
Transec											
1		780	1542	559	559	200	200	1948	2436	0	0
2		722	900	1047	1047	484	484	2159	1740	199	200
3		567	567	1051	1051	636	636	956	480	0	0
4		232	232	1102	1102	502	502	531	397	138	145
5		269	269	1226	1226	886	886	1350	971	0	0
6		508	508	874	874	702	702	1296	778	153	288
7		195	195	219	219	363	363	290	246	125	181
8		0	0	0	0	0	479	3540	3714	298	308
9		0	0	0	0	0	554	4165	3258	318	331
10		0	0	0	0	0	620	1436	852	0	0
11		0	0	0	0	0	812	3017	1880	513	548
12		0	0	0	0	0	1032	2124	1498	509	616
13		0	0	0	0	0	734	2591	1520	481	459
14		0	0	0	0	0	320	448	339	262	251
15		0	0	0	0	0	655	1316	982	278	314
16		0	0	0	0	0	1076	2750	1959	0	0
17		0	0	0	0	0	497	1529	1070	414	394
18		0	0	0	0	0	1013	3723	2177	507	526
Grand Total		3273	4213	6078	6078	3773	11565	35169	26297	4195	4561

lake	Henderson																
ats year																	
Tot Targets																	
r (sequential dates)																	
2000-07-11		2000-07-12		2004-03-09		2004-03-10		2004-11-28		2004-11-29		2004-11-30		2007-08-23		2007-08-24	
TARGET00.DAT		TARGET03.DAT				TARGET04.DAT				TARGET07.DAT							
Rankin - Hanslitt Survey		Rankin - Hanslitt Survey		processed by pr, 14May2004				processed by BH, 06Dec2004				processed by SC, 13Sep2007.					
clear to cloudy,rippled to chop, 3/4 moon many transects replicated with EY500. PR counter.		12Jul00/22Jan 01/BH/EY500/0/0/0/17/225/15		clm,clr,fltmn,A T4WT6@2300 7/37/104;unadjusted for size bias		clm,clr,fltmn,A T4WT6@2300 7/37/104;adjusted for size-bias		Corrected for size bias 30Nov04/06Dec04/BH/EYM/2 0069/0/0/0/2/9 0/24		30Nov04/06Dec04/BH/EYM/2 0069/0/0/0/2/8 0/24		30Nov04/06Dec04/BH/EYM/2 0069/0/0/0/2/90/24		clear, calm		clear, calm	
Transec																	
1		69	74	403		403		153	153	153		0					0
2		219	131	689		689		414	414	414		0					0
3		269	238	565		565		483	483	483		250				250	
4		230	262	489		489		367	367	367		94				94	
5		323	347	694		694		390	390	390		115				115	
6		400	428	1186		1186		485	485	485		140				140	
7		244	217	112		112		356	356	356		138				138	
8		219	0	935		935		778	778	778		0				0	
9		461	0	1813		1813		900	900	900		712				712	
10		485	0	2379		2379		413	413	413		702				702	
11		828	0	1599		1599		944	944	944		388				388	
12		1022	0	2311		2311		1049	1049	1049		499				499	
13		620	0	535		535		530	530	530		333				333	
14		257	250	148		148		166	166	166		78				78	
15		0	0	251		251		360	360	360		354				354	
16		0	0	366		366		414	414	414		0				0	
17		656	0	556		556		569	569	569		0				0	
18		972	0	1955		1955		1278	1278	1278		0				0	
Grand Total		7274	1947	16986		16986		10049	10049	10049		3803				3803	

- 86-09-23 – deleted, as 86-09-24 has corrections
- 93-02-25 – deleted, as 93-02-26 has corrections to species proportions
- 93-07-16 – deleted, as 93-07-17 was based on all transects
- 94-06-26 and 94-06-27 are assumed to be bona fide replicates due to differences in methodology
- 95-06-20 – deleted, as 95-06-21 appears to be the revised version (i.e., more metadata associated with it)
- 2000-07-11 and 2000-07-12: here the nature of the differences are unclear, therefore all data retained
- 2004-03-09 – deleted, as 2004-03-10 has adjustments for size bias
- 2004-11-29 and 2004-11-30 – both deleted, as 2004-11-28 has adjustments for size bias

- **2007-08-23** / 2007-08-24 – differs in species proportions for some transects/depths, but unclear which version is the ‘adjusted’ version. **2007-08-23 deleted.**

Heydon Lake

lake	Heydon Lk	
ats_year	(All)	
Tot Targets	Survey	
	Replicate or potential	
	2000-11-09	2000-11-10
	JUVENILE	
	SIMRAD	
	2	
	TARGET00.DAT	
	Simpson-Murry Survey	Simpson-Murray Survey
	Juvenile Targets Only - Trans #6 Used	Juvenile targets only - Trans 6 Not Used
Transec		
1	93	93
2	128	128
3	132	132
4	205	205
5	162	162
6	262	0
7	0	0
8	0	0
Grand Total	982	720

- **2000-11-10 deleted**, as incomplete, according to comments

Hobiton Lake

like ats year	Hobilton Lk [All]	▼																							
Tot Targets		Survey ▼																							
		1987-04-20	1987-04-21	1992-08-06	1992-08-07	1992-09-01	1992-09-02	1992-09-03	1992-09-04	1993-07-14	1993-07-15	1993-08-11	1993-08-12												
		Replicate or potential duplicate (s)																							
														JUVENILE											
														SIMRAD											
														2											
		TARGET87.DAT				TARGET92.DAT				TARGET93.DAT															
		NA		processed by PR, 10Aug92.		Counted Aug.92, SM		Counted Sept.3/92 SM		processed by PR, 18Jan92.		processed by PR, 18Jan93.		Counted Aug.5/93 SM		processed by PR, 24Aug93.		Counted Sept.8/93 SM							
		100% cloud til 2400, no wind, precip, or moon, AT= No stickies in trawl catch. Re output: prop. of 0= sockeye given as % Sc.		100% cloud til 2400, no wind, precip, or moon, AT= No stickies in trawl catch. Re output: prop. of 0= sockeye given as % Sc.		overcast, light precip, AT 15 ST 20 deg C. EY M #10005. Sounder # not recorded, assumed most frequently used sounder.		ovrcst, light precip, AT 15, ST 20 deg C. EY- M #10005. Sounder not recorded, assumed most frequently used sounder.		@0136, 100% cid, fog, 0 precip, calm, 0moon, AT 13, EYM 20111, gain 10, 40 log scale, dynaline ON.		@0136, 100% cid, fog, 0 precip, calm, 0moon, AT 1 EYM 10005, gain 10, 40 log scale, Dynaline ON.		EYM #10005		EYM #20111		ovcst, ripple, driz., at 13.0 wt 18.0 0moon this set of transects performed by SM		ovcst, ripple, driz., at 13.0 wt 18.0, 0moon transects performed by AI Hirst EYM 20111.		foggy, calm, AT14 ST 20.5 degc. #20111.		fill in later EY- M 20111 gain 10 log 40 dynaline on.	
Transect ▼																									
1		42		42		24		26		60		96		83		53		120		101		64		69	
2		24		24		7		8		56		53		49		53		143		163		139		184	
3		98		98		100		103		90		54		44		81		43		43		76		77	
4		148		148		294		334		319		420		327		259		50		37		61		63	
5		42		42		375		385		252		284		242		188		161		177		57		61	
6		108		108		29		33		128		178		165		111		206		197		183		210	
7		0		0		37		36		107		164		137		80		523		636		408		479	
8		332		0		237		256		242		280		251		209		545		596		431		508	
9		196		196		289		307		215		219		184		178		93		93		112		114	
10		329		0		355		378		352		449		366		278		78		65		121		124	
11		163		163		387		379		454		561		451		216		85		80		111		111	
Grand Total		1482		821		2134		2245		2275		2738		2299		1706		2047		2188		1763		2000	

[illegible]

- 87-04-20 & 87-04-21 – latter assumed incomplete, deleted
- 1992, 1993, 1994, 1995 surveys – different methods or processors – no removals because it's unclear which of the replicates are correct.

Ian Lake

lake	Ian Lk						
ats_year	(All)						
Tot Targets	Survey						
	Replicate or potential duplicate (sequential survey dates)						
	1994-09-06	1994-09-07	1994-09-08	1995-09-05	1995-09-06	2003-09-07	2003-09-08
	JUVENILE						
	SIMRAD						
	2						
	TARGET94.DAT			TARGET95.DAT		TARGET03.DAT	
	EYM#20111,DAT#A61400			NA		Hanslit-Katinic Survey	
Transec	c.c.70%,winds light,waves 10cm A.T.13.3,H2O 17.0,no moon. Processed by S.T.	c.c.70%,winds light,waves 10cm A.T.13.3,H2O 17.0,no moon. Processed by S.T., species reassigned by pr, 22Feb95.	c.c.70%,winds light,waves 10cm A.T.13.3,H2O 17.0,no moon.Process ed by S.T.Tr 8&9 zeroed for comparison, 30Aug96	05Sep95/24No v95/ST/EYM/20 069//0/0/0/9/1 1/0/	Transects 1 & 2 "zeroed" for statistical compariso 05Sep95/24No v95/ST/EYM/20 069//0/0/0/9/1 1/0/	7Sep03/15Oct 03/BH/EY500/0 /0/0/14/0/0	Uses 12Sep1996 proportions.
1	15	15	15	17	17	19	19
2	22	22	22	26	26	40	40
3	46	46	46	60	60	59	59
4	34	34	34	42	42	94	94
5	99	99	99	118	118	156	156
6	123	123	123	138	138	98	98
7	30	30	30	47	47	4	4
8	169	169	169	163	163	0	0
9	142	142	142	156	156	0	0
Grand Total	680	680	680	767	767	470	470

- 1994, 1995 surveys – different species compositions, therefore all records retained
- 2003 surveys – different species proportions, no preference given, 2003-09-07 deleted assuming 2003-09-08 has a fix

Kennedy Lake (Clayoquot Arm)

lake	Kennedy Lk (Clayoquot Arm)									
ats_year	(All)									
Tot Targets	Survey									
	1985-01-25	1985-01-26	1985-04-02	1985-04-03	1985-05-31	1985-06-01	1985-06-28	1985-06-29	1985-06-30	1985-07-01
	FURUNO FM 22					SIMRAD		FURUNO FM 22		
	1					2		1		
	Juvenile Survey			Juvenile survey, 2 nights	Juvenile survey, 4 nights	Juvenile Survey	Juvenile survey, 3 nights			
Transec	100%, no ppt, calm, 3/4 moon, Wt: 14.5 Oc, At: 4.5 Oc (0000) JC 851022.	variable 0-100%, calm, no moon, no ppt, At: 6/9.5 Oc (1830), Wt: 6 Oc, PR 850205.	0% (fog), light drizzle, full moon, calm, At: 6 Oc At: 6.5/7.0 Oc (2000), Wt: 6 Oc, JC 061285.	clear->cloudy, calm, full moon, drizzle 2nd night; trans 1, 10-20m; trans 2 & 3, PR COUNTER	100->10% cloud, calm, >rippled, 3/4-full moon ST: 14.5 deg C, PR COUNTER	10-100%, calm, no ppt, 3/4 moon, Wt: 14.5 Oc, JC 851202	clear, light breeze, 3/4 moon; Wt: 20 Oc, At: 15/14 Oc (2300/2300) Trans 1, 5-10m, min. count, PR COUNTER	0%, breeze, no ppt, 3/4 moon, At: 15/14 Oc (2300), Wt: 20 Oc, JC 851129	clear, light breeze, 3/4 moon, AT 15 @2300hr, ST 2 Furuno #7713, PR COUNTER.	0%, breeze, no ppt, 3/4 moon, At: 14/15 Oc (2300), Wt: 20 Oc, JC 851126.
1	1449	1018	1887	1564	131	131	637	657	271	263
2	4699	3957	4590	3697	310	317	1192	1027	481	469
3	2195	1888	3269	2682	160	174	761	639	229	276
4	1067	680	1133	1073	168	149	431	536	129	157
5	1516	1098	1060	979	106	113	499	557	115	132
6	808	630	620	505	120	127	356	382	88	83
7	336	287	269	240	122	127	159	174	32	37
8	0	0	0	0	0	0	0	0	0	0
Grand Total	12070	9556	12828	10740	1117	1138	4035	4172	1345	1417

lake	Kennedy (Clay)									
ats_year										
Tot Targets										
Replicate or potential duplicate (sequential survey dates)										
	1985-07-10	1985-07-11	1985-07-12	1985-07-13	1985-08-13	1985-08-14	1985-08-15	1985-08-16	1985-08-17	
	JUVENILE									
	SIMRAD		FURUNO FM 22		SIMRAD		FURUNO FM 22			
	2		1		2		1			
	TARGET85.DAT									
	Juvenile Survey									
	clear, rippled, 1/2-1/4 moon by 0315, Wt: 20.5 Oc At: 17.5 Oc (2315), PR counter.	0%, light winds, no ppt, 1/4-1/2 moon up by 0315, At: 17.5 (2315), Wt: 20 Oc, JC 851203	0%, light winds, no ppt, 1/4-1/2 moon up by 0315, At: 17.5 Oc (2315), Wt: 20 Oc, PR counter, Furuno #7713	0%, wind light, no ppt, 1/4-1/2 moon, At: 17.5 Oc (2315), Wt: 20 Oc, JC 851126.	clear, calm, no moon; AT 24/17/10 Oc (2300/2300/0000) Wt: 20 Oc, Tr 1 & 6, 5-10 m strata are minimum counts, PR COUNTER.	0%, calm, no ppt, no moon, At: 24/17/18 Oc (2330/2330/0000), Wt: 20 Oc, JC 851129.	clear, calm, no moon; AT 24/17/18 Oc, (2330/2330/000) Wt: 20 Oc, FM-22 #7713, PR counter.	0%, calm, no ppt, no moon, At: 24/17/18 (2330/2330/0000) Oc, Wt: 20 Oc, JC 851126	0%, calm, no ppt, no moon, At: 24-17 Oc, (3 nites) Wt: 20 Oc, JC count #2	
Transec										
1	514	548	253	283	456	681	268	305	1046	
2	519	557	215	240	688	819	406	438	1079	
3	423	476	194	211	434	536	275	288	1111	
4	507	606	244	244	398	477	252	254	950	
5	689	693	286	290	1109	1410	821	756	1533	
6	560	585	193	192	1022	1294	760	797	1300	
7	240	249	62	66	473	499	295	288	770	
8	0	0	0	0	0	0	0	0	0	
Grand Total	3452	3714	1447	1526	4580	5716	3077	3126	7789	

lake	Kennedy (Clay)									
ats_year	(Ctrl)									
Tot Targets										
	1985-09-17	1985-09-18	1985-10-30	1985-10-31	1989-09-28	1989-09-29	1989-09-30	1995-07-25	1995-07-26	
	SIMRAD									
	2									
					TARGET89.DAT		TARGET95.DAT			
					processed by RT on Nov. 8'89.	processed on 19Dec89 by PR.	processed by JC	NA	Processed by J.H. Aug11/95	
Transec	clear, light breeze; AT 13 Oc (0000), Wt: 17 Oc, trans 6, 5-10 m strata, min. count, PR COUNTER.	0%, light breeze, no ppt, no moon, At: 13 Oc (0000) Wt: 17 Oc, JC 851209.	overcast, gusting wind, rain, At: 6.5/11 Oc (2330/0000), PR counter.	85-100%, gusty, heavy rain, moon full obscured, At: 6.5/11 Oc (2330/0000), Wt: 10 Oc, JC 851206.	10-100% cldcvr, 2d, AT14.5ST19 8KS.W.wmd, at 14.5, st19C@ 2346	ovrcst, SWwin d, AT14.5ST19 degC@2346. processed by PR, 19Dec89.	10-100% cldcvr, a t14.5, st19C 7 trawls: 60 0+ Sx; 121 Stkb.	26Jul95/120c 195/yr/eym/2 0128/1/0/1/ 23/2/316	air 16.0, h2o 20.0, wind light, waves 3cm, rain no 26Jul95/11A 1U095/JH/EY M/20128/1/0 /0/23/2/316	
1	756	813	1015	1546	1049	1236	1238	211	210	
2	1023	1239	1474	1870	1607	1621	1585	593	590	
3	465	501	1282	1543	1493	1712	1889	360	352	
4	350	388	822	1011	1130	1131	1168	190	181	
5	913	1042	652	815	3527	3559	4012	658	576	
6	1680	1886	687	848	3036	3178	3995	258	243	
7	546	617	430	503	1382	1478	1493	235	240	
8	0	0	0	0	0	0	0	0	0	
Grand Total	5733	6486	6362	8136	13224	13915	15380	2505	2392	

- All surveys displayed appear to be bona fide replicate field surveys or different lab processor/methods. No changes advised.

Kennedy Lake (Main Arm)

lake	Kennedy Lk (Main Arm)																							
ats_year	All																							
Sum of targets	Survey Date																							
	1985-12-11	1985-12-12	1986-09-18	1986-09-19	1986-10-07	1986-10-08	1986-10-09	1986-11-19	1986-11-20	1986-11-21	1987-04-23	1987-04-24	1993-10-20	1993-10-21	1993-10-22									
	JUVENILE																							
	SIMRAD		FURUNO FM 22		SIMRAD		FURUNO FM 22		SIMRAD		FURUNO FM 22		SIMRAD		FURUNO FM 22									
	TARGET85.DAT		TARGET86.DAT		TARGET87.DAT		TARGET88.DAT		TARGET89.DAT		TARGET90.DAT		TARGET91.DAT		TARGET92.DAT									
	0%, calm, no moon, At: 4.0 Oc, Wt: 6.0 Oc JC 851216.		5%, calm, no ppt, full moon by 2100, At: 16/13 Oc (2020/0138), Wt: 18 Oc, JC 861016, targ overlap, Juvenile.		5%, calm, no ppt, full moon by 2100, At: 16/13 Oc (2020/0138), Wt: 18 Oc, JC 861021, Juvenile.		0%, fog, no ppt, mod strong S wind, 1/4 moon till At: 12/14 Oc (2030/0020), JC 861008, corrected for target overlap 18/JAN89 PR		0%, fog, no ppt, mod strong S wind, 1/4 moon till At: 12/14 Oc (2030/0020), Wt: 15 Oc, JC 861010, targ overlap, Juvenile.		0%, fog, no ppt, strong S wind, 1/4 moon till At: 12/14 Oc (2030/0020), Wt: 15 Oc, JC 861010, JC870128, poor targ signal, Juvenile.		100%, gusts, showers, 3/4 moon throu cloud by 0000 At: 13/10 (2300/0415), Wt: 9.0 Oc, JC870128, poor targ signal, Juvenile.		100%, gusts, showers, 3/4 moon vis through cloud, At: 13/10 Oc (2300/0415), Wt: 9.0 Oc, corrected for targ overlap 18/JAN89 PR		100% cloud, no moon, intermittent wind & precip AT=12deg, ST=11deg, small areas of target overlap on several transects.		100% clds,intermittent rain@wind, air@1836 = 12.0, wat transect #3,#4,#5,@ #7 out by 145,144,175,@258.		20%cld,light-mod wind, AT ?? WT ?? This estimate uses additional trawl data which reduces the proportion of sockeye.		20%cld,light-mod wind, 0-30 cm waves, AT 6.0 wt 15 Separated transect 1 from other transects when calculating spp. comp.	
Transect	Juvenile Survey	Juvenile Survey	Juvenile Survey	Juvenile Survey	Juvenile Survey	Juvenile Survey	Juvenile Survey	Juvenile Survey	Juvenile Survey	Juvenile Survey	NA	Juvenile Survey	processed by PR, 08Nov93.	Counted by PR	Counted by P.R. Nov.8/93									
1	4443	1219	1060	2222	3721	2971	1575	1696	2872	3932	734	740	1495	1495	1495									
2	4000	1821	911	2065	2484	1594	907	1487	2774	3434	1407	1367	1792	1792	1792									
3	2767	1410	709	1450	2874	2874	1101	1226	1450	1450	1263	1343	1632	1632	1632									
4	2390	1105	940	1832	2843	2843	1268	1263	2286	3016	800	766	1129	1129	1129									
5	1143	484	1093	1552	2625	2625	1164	541	950	950	983	995	943	943	943									
6	71	37	945	1550	5356	3126	1225	383	819	819	455	481	336	336	336									
7	114	51	884	1881	1565	1565	342	262	415	415	332	351	1228	1228	1228									
Grand Total	14928	6127	6542	12552	21468	17598	7582	6858	11566	14016	5974	6043	8555	8555	8555									

lake	Kennedy Lk (Main Arm)			
ats_year				
Sum of targets	1995-07-24	1995-07-25	2000-10-24	2000-10-25
	TARGET95.DAT		TARGET00.DAT	
	air 19.0, H2O 20.0, wind light, waves 15cm, rain h 24JUL95/08 AUG95/JH/ EYM/10005 /20128/1/0 /0/21/0/14 3	24Jul95/12 Oct95/pr/e ym/10005/ 20128/21/0 /143	A minimum value of .01 was given to the 1 sk that 24Oct00/25J an01/BH/EY5 00/0/0/0/15/ 1/298	24Oct00/31J an01/BH/EY M/20069/0/0 /0/15/1/298
	Processed by J.H. Aug 8/95	NA	Hanslit Survey	Hanslit Survey
1	477	531	2655	2738
2	527	547	3442	2826
3	706	756	2838	2543
4	279	304	1959	1498
5	362	375	1673	1398
6	968	968	912	826
7	686	825	0	494
Grand Total	4005	4306	13479	12323

- 93-10-20 & 93-10-21 – deleted, superseded by 93-10-22
- In the lower table, assumed bona fide replicates.

Long Lake (A)

lake	Long Lk (A)					
ats_year	(All)					
Tot Targets	Survey					
	Replicate or potential duplicate (sequential survey dates)					
	2007-02-19	2007-02-20	2007-11-07	2007-11-08	2008-02-22	2008-02-23
	JUVENILE					
	SIMRAD					
	2					
	TARGET06.DAT		TARGET07.DAT			
	Hanslit-Bachen Survey		processed by RF March 08		NA	
Transec	19Feb07/8Mar07/EYM /20101/BH/0/0/0/13/7 /116/183	19Feb07/8Mar07/EYM /20101/BH/0/0/0/13/7 /116/183	depth 0-120, no STR	depth 0-120, no STR Adjusted	clear, calm survey by RF and MM	clear, calm survey by RF and MM, adjusted
1	116	116	175	175	79	79
2	0	0	224	224	171	171
3	138	138	367	367	291	291
4	430	430	477	477	1205	1205
5	621	621	459	459	2039	2039
6	0	0	0	0	0	0
7	453	453	582	582	990	990
8	1261	1261	1754	1754	1799	1799
Grand Total	3019	3019	4038	4038	6574	6574

- 2007-02-19 – appears identical to 2007-02-20, i.e., no changes in targets or proportions; **2007-02-19 deleted**
- **2007-11-07 – deleted**, given 2007-11-08 is identical but with adjusted proportions
- **2008-02-22 – deleted**, given 2008-02-23 is identical but with adjusted proportions

Long Lake (B)

lake	Long Lk (B)					
ats_year	(All)					
Tot Targets	Survey					
	Replicate or potential duplicate (sequential survey dates)					
	2007-02-19	2007-02-20	2007-11-07	2007-11-08	2008-02-22	2008-02-23
	JUVENILE					
	SIMRAD					
	2					
	TARGET06.DAT		TARGET07.DAT			
	Hanslit-Bachen Survey		processed by RF Jan 08		processed by RF	
Transec	19Feb07/8Mar07/EYM /20101/BH/0/0/0/13/5 6/45	19Feb07/8Mar07/EYM /20101/BH/0/0/0/13/5 6/45	0-120 depth setting, no weather report	0-120 depth setting, no weather report Adjusted	clear, calm	clear, calm adjusted
1	359	359	683	683	156	156
2	581	581	637	637	321	321
3	1573	1573	684	684	1246	1246
4	688	688	40	40	1733	1733
5	981	981	2049	2049	854	854
6	2408	2408	987	987	1865	1865
7	1458	1458	567	567	2424	2424
Grand Total	8048	8048	5647	5647	8599	8599

- 2007-02-19 – appears identical to 2007-09-20, i.e., no changes in targets or proportions; **2007-02-19 deleted**
- **2007-11-07 – deleted**, given 2007-11-08 is identical but with adjusted proportions
- **2008-02-22 – deleted**, given 2008-02-23 is identical but with adjusted proportions

Mercer Lake

lake	Mercer Lk	
ats_year	(All)	
Tot Targets	Survey	
	Replicate or potential duplicate (sequential)	
	1986-08-14	1986-08-15
	JUVENILE	
	SIMRAD	
	2	
	TARGET86.DAT	
	Juvenile Survey	Mercer Test
Transec	40%, light winds, no ppt, no moon, At: 12/10 0c (2230/0300), Wt: 14.5 0c, JC 861003, Juvenile,	this is a test to compare differences in population estimates when the estimate is run up using the old transects 1-4 only.
1	303	303
2	363	363
3	238	238
4	68	68
5	471	0
6	645	0
Grand Total	2088	972

- 86-08-14 vs. **86-08-15** – a comparison in pop estimates with two transects excluded, may be useful for calibrating surveys based only on transects 1-4, although that kind of analysis seems doable with any and all surveys that feature all six transects. Note that 860814 is more complete, so **86-08-15 was the logical deletion**.

Meziadin Lake

lake	Meziadin Lk	
ats_year	(All)	
Tot Targets	Survey	
	Replicate or potential duplicate (sequential)	
	1991-08-30	1991-08-31
	JUVENILE	
	SIMRAD	
	2	
	TARGET91.DAT	
	counted Nov26/91 SM	counted Nov26/91
Transec	70% cld, calm, 2/3 moon 2230-?, AT 8.0, Wt 11.0, @ 0310 Bottom noise transects 1,2,3,4 & 5.	70% cld, calm, 2/3 moon 2230-?, AT 8.0, Wt 11.0, @ 0310 Bottom noise transects 1,2,3,4 & 5.
1	0	144
2	0	413
3	0	883
4	0	866
5	0	987
6	0	125
Grand Total	0	3418

- 91-08-30 – deleted**, as it contains only zeros for targets (and spp. composition).

Muchalat Lake

lake	Muchalat Lk	
ats_year	(All)	
Tot Targets	Survey	
	Replicate or potential duplicate (sequential	
	1995-08-14	1995-08-15
	JUVENILE	
	SIMRAD	
	2	
	TARGET95.DAT	
	NA	
Transec	c.c.50%,winds calm,waves 0cm,ST18.5 14Aug95/07Sep95/ST/ EYM/20128//1/0/0/6/2 81/0	14Aug95/26Mar96/pr/ eym/20128//1/0/1/6/2 81/0
1	339	254
2	643	532
3	259	216
4	515	379
5	727	382
Grand Total	2483	1763

- 95-08-14 and 95-08-15 – different processors; no changes required

Muriel Lake

lake	Muriel Lk						
ats_year	(All)						
Tot Targets	Survey						
	Replicate or potential duplicate (sequential survey dates)						
	1986-11-03	1986-11-04	1986-11-05	1989-09-26	1989-09-27	2000-10-30	2000-10-31
	JUVENILE						
	SIMRAD						
	2						
	TARGET86.DAT			TARGET89.DAT		TARGET00.DAT	
	Juvenile Survey			processed by JC	Counted by RT on Nov.8'89	Hanslit Survey	
Transec	100%, calm, light breeze, no ppt, no moon, Transition Survey run during dusk 1630-1815, Juvenile.	100%, light breeze, no ppt, no moon, JC 861105, Night Survey 2045-2230, no spp break-down, Juvenile.	100%, light breeze, no ppt, JC 861105, This is a day survey 0915- 1100, Juvenile.	10- 100%cldcvr,at 15.5, st19C above 5m: .9- 0+,.05-1+. processed by JC	10- 100%cldcvr,W wnd0- 5knts,at15.5,st 19C@2225 above 5m: .9- 0+,.05-1+. Counted by RT on 08Nov89.	30Oct00/21De c00/BH/EYM/2 0069/0/0/0/7/1 38	Hanslit Survey 31Oct00/16Jan 01/BH/EY500/0 /0/0/7/138/0
1	9	37	0	13	10	30	37
2	73	118	57	28	26	135	82
3	88	130	80	40	37	145	101
4	61	200	61	129	120	276	127
5	140	219	108	112	114	385	256
6	107	180	121	81	82	311	245
7	100	184	94	114	117	217	182
8	116	206	78	150	145	276	209
9	63	98	54	43	44	121	72
10	45	52	2	21	19	33	38
Grand Total	802	1424	655	731	714	1929	1349

- 86-11-03 and -04 and -05 – comparative surveys at different times of day; no changes required
- 89-09-26 and -27 – different processors; no changes required
- 2000-10-30/31 – appear to be different readings on different dates; no changes advised

Nimpkish Lake

lake	Nimpkish Lk	▼
ats_year	(All)	▼
Tot Targets	Survey	▼
	Replicate or potential duplicate (sequential	
	1989-09-13	1989-09-14
	JUVENILE	
	SIMRAD	
	2	
	TARGET89.DAT	
	processed by JC	
Transec	clr,flmn,at13,st16.5, mnup@2015 WARNING-40 fish in surface trawls. 12trwls: Sx 419-0+,1- 1+,Stkb 31. processed by JC	see Nimpkish Lk - 890913 40 fish in surface trawl. Recount by J Cuell, 01Dec89.
1	1217	915
2	4247	3522
3	3771	3223
4	6550	4040
5	3757	3104
6	3616	3535
7	4046	3078
8	686	487
Grand Total	27890	21904

- 89-09-13 and -14 – different processors; no changes required

Phillips Lake

lake	Phillips Lk	▼
ats_year	(All)	▼
Tot Targets	Survey	▼
	Replicate or potential duplicate (sequential	
	survey dates)	
	1986-08-21	1986-08-22
	JUVENILE	
	SIMRAD	
	2	
	TARGET86.DAT	
	Phillips test	Juvenile Survey
Transec	This is the Phillips estimate using the original transects	0%, calm, no ppt, At: 16 0c, Wt: 17 0c, full moon by 2300, targ overlap at head of lake, Juvenile.
1	982	982
2	336	336
3	147	147
4	29	29
5	97	97
6	0	962
7	0	286
8	0	178
9	0	100
Grand Total	1591	3117

- 86-08-21 vs 86-08-22 – a comparison in pop estimates with four transects excluded, may be useful for calibrating surveys based only on transects 1-5, though that kind of analysis seems doable with any and all surveys that feature all nine transects. Note that 86-08-22 is more complete, so **86-08-21 was the logical deletion**.

Osoyoos Lake (N)

lake	Osoyoos LK (▼)										
ats_year	(All) ▼										
Tot Targets	Survey ▼										
	⊖ 1998-10-02	⊖ 1998-10-03	⊖ 2000-07-05	⊖ 2000-07-06	⊖ 2000-09-27	⊖ 2000-09-28	⊖ 2004-03-02	⊖ 2004-03-03	⊖ 2005-08-04	⊖ 2005-08-05	
	JUVENILE										
	⊖ TARGET98.DAT			⊖ TARGET00.DAT			⊖ TARGET03.DAT		⊖ TARGET05.DAT		
	SIMRAD										
	2										
	⊖ Data collection with EY500	⊖ processed by PR,12Jan98.	⊖ processed by PR, 09Aug2000.	⊖ Hanslit - ONFC Survey	⊖ Hanslit Survey		⊖ processed by pr, 07jun2004		⊖ Rankin Survey		
	EP500 run to obtain target counts	Transects 1 & 5, Other transects done using ey500.	100-70%cld/norn/hghwnd/AT15/ST19/05jul00/09aug00/pr/eym/20069/0/0/0/10/486/0	5Jul00/18Jan01/BH/EY500/0/0/0/10/486/0	40-50m strata counts added to the 30-40m strata 27Sep00/21Dec00/BH/EYM/20069/0/0/0/10/331/0	40-50m strata counts added to the 30-40m counts 27Sep00/17Jan01/BH/EY500/0/0/0/10/331/0	6/138/9-1+	Adjusted for size bias,6/138/9-1+	NA	Target count totals corrected by PR, 15Jun2006.	
Transec ▼											
1	0	3337	447	576	1358	1316	289	289	1048	1377	
2	4477	0	1130	1492	1187	1511	729	729	3654	3521	
3	3300	0	1142	1859	623	697	324	324	2317	2217	
4	5028	0	1111	0	1013	1696	616	616	1281	1281	
5	0	2335	522	793	1194	1198	249	249	1459	1466	
6	2829	0	718	833	842	926	333	333	1362	1362	
7	4821	0	1805	2019	1671	1712	732	732	4456	4359	
8	2018	0	654	1120	459	506	282	282	2604	2604	
9	4662	0	656	1167	1106	1274	390	390	1071	1071	
10	8019	0	612	0	1541	1178	258	258	2092	2092	
Grand Total	35154	5672	8797	9859	10994	12014	4202	4202	21344	21350	

- 98-10-02, and -03 – complementary transect data from different methods (EP500 vs EY500); presumably 98-10-03 was focused on transects 1 and 5 to fill in the blanks in 98-10-02, but that is not clearly stipulated. Both datasets retained.
- 2000-07-05/06 – different processors
- 2000-07-27/28 – different count analyses though same processor
- 2004-03-02/03 – 04-03-02 deleted as 04-03-03 species proportions were adjusted for size bias
- 2005-08-04/05 – 05-08-04 deleted as 05-08-05 target counts were corrected

Owiken Lake (A)

lake	Owiken Lk (A)					
ats_year	All					
Tot Targets	Survey Date					
	2004-02-04	2004-02-05	2007-02-14	2007-02-15	2008-02-20	2008-02-21
	JUVENILE					
	SIMRAD					
	TARGET03.DAT		TARGET06.DAT		TARGET07.DAT	
	4Feb04/23Mar07/EYM/20069/0/0/20/937/217	Corrected for size bias. 4Feb04/23Mar07/EYM/20069/0/0/20/937/217	14Feb07/22Mar07/BH/EYM/20101/0/0/19/604/154		clear, calm	clear, calm adjusted
Transect	Hanslit-Bachen Survey	Hanslit-Bachen Survey	Hanslit-Bachen Survey	Hanslit-Bachen Survey	processed by RF	processed by RF
1	3232	3232	755	755	331	331
2	515	515	266	266	127	127
3	650	650	826	826	66	66
4	216	216	620	620	34	34
5	47	47	219	219	12	12
6	38	38	0	0	1	1
7	4180	4180	0	0	4214	4214
8	82	82	4021	4021	3706	3706
9	544	544	357	357	1147	1147
10	492	492	347	347	694	694
11	674	674	642	642	123	123
12	180	180	435	435	44	44
13	62	62	200	200	9	9
Total	10912	10912	8688	8688	10508	10508

- 2004-02-04 – **deleted**, as 2004-02-05 adjusted for size bias
- 2007-02-14/15 – data contain different species proportions without indication of rationale (size bias?) or which survey was adjusted (likely 070215). **2007-02-14 deleted.**
- 2008-02-20 – **deleted**, as 2008-02-21 adjusted for size bias

Owiken Lake (B)

lake	Owiken Lk (B)					
ats_year	(All)					
Tot Targets	Survey					
	Replicate or potential duplicate (sequential survey dates)					
	2004-02-04	2004-02-05	2007-02-14	2007-02-15	2008-02-19	2008-02-20
	JUVENILE					
	SIMRAD					
	2					
	TARGET03.DAT	TARGET06.DAT	TARGET03.DAT	TARGET06.DAT	TARGET07.DAT	
	Hanslit-Bachen Survey				processed by RF	
	4Feb04/23Mar07/EYM/20069/0/0/20/937/217	4Feb04/23Mar07/EYM/20069/0/0/20/937/217	4Feb04/23Mar07/EYM/20069/0/0/20/937/217 Corrected for size-bias.	14Feb07/22Mar07/BH/EYM/20101/0/0/19/604/154	14Feb07/22Mar07/BH/EYM/20101/0/0/19/604/154	clear, calm clear, calm adjusted
Transect						
1	6370	6370	6370	658	658	4593
2	1751	1751	1751	816	816	2815
3	1524	1524	1524	379	379	1480
4	1284	1284	1284	1636	1636	550
5	3916	3916	3916	993	993	3851
6	1469	1469	1469	939	939	888
7	435	435	435	1915	1915	498
8	270	270	270	1014	1014	512
Grand Total	17019	17019	17019	8350	8350	15187

- 2004-02-04 – **and copy from TARGET06.DAT: deleted**, as 2004-02-05 adjusted for size bias
- 2007-02-14 and -15 – data contain different species proportions without indication of rationale (adjusted for size bias?) or which survey was adjusted (likely 07-02-15). **2007-02-14 deleted.**
- 2008-02-19 – **deleted**, as 2008-02-20 adjusted for size bias

Sakinaw Lake

lake	Sakinaw Lk	
ats_year	(All)	
Tot Targets	Survey	
	Replicate or potential duplicate (sequential survey dates)	
	2002-09-16	2002-09-17
	JUVENILE	
	SIMRAD	
	2	
	TARGET02.DAT	
	processed by bh, 19Sep02.	
	clear,,831mn 16Sep02/19Sep02/EY M/20069/bh/0/0/0/5/4 6/84.	clear,,831mn 16Sep02/19Sep02/EY M/20069/bh/0/0/0/5/4 6/84.
Transec		
1	37	37
2	20	20
3	183	183
4	113	113
5	71	71
6	81	81
7	70	70
8	88	88
9	179	179
10	163	163
11	27	27
12	382	382
13	394	394
14	225	225
Grand Total	2033	2033

- 2002-09-16/17 – survey data have different species proportions, without notes on which survey is correct, assume the second one is correct. 2002-09-16 deleted.

Skaha Lake

lake	Skaha Lk					
ats_year	(All)					
Tot Targets	Survey					
	Replicate or potential duplicate (sequential survey dates)					
	2000-07-08	2000-07-09	2000-09-30	2000-10-01	2004-05-11	2004-05-12
	JUVENILE					
	SIMRAD					
	2					
	TARGET00.DAT			TARGET04.DAT		
	processed by PR, 010900.	Hanslit - ONFC Survey		Hanslit Survey # INVALID DATE (00/09/31) - RESET to 09/10/01 in	processed by pr, 10Aug2005.	
	100%cid/trn/hghwnd/ AT12/ST18/ 08jul00/01sep00/pr/e ym/20069/no trawls.	9Jul00/22Jan01/BH/ EY500/0/0/0/NO Trawling	30Sep00/21Dec00/B H/EYM/20069/0/0/0/ 7/9/0	30Sep00/16Jan01/B H/EY500/0/0/0/7/9/0 # REPLICATE SURVEY DATE? #	30pctcid,lightNE,A14; ST9.3deg Small & large targets	30%cid;lightNEwmd;A T4;ST9.3degC. Small targets only.
Transec						
1	358	392	510	524	323	208
2	188	283	293	490	1043	601
3	309	331	429	445	1041	613
4	60	65	35	18	392	318
5	108	51	214	147	0	0
6	0	0	0	0	1065	684
7					584	396
8					0	0
9					0	0
Grand Total	1023	1122	1481	1624	4448	2820

- 2000-07-08/09 – different software (EYM vs EY500?); no changes advised
- 2000-09-30/10-31 – different software (EYM vs EY500?); no changes advised
- 2004-05-11/12 – different target size inclusion; no changes advised

Sproat Lake

[illegible]

- 1989 & 1990 and 95-07-13 and -14 surveys: different processors; no removals
- 95-11-20 and -21 – different species proportions, but no final preference specified; deleted 95-11-20

Skidegate Lake NE

lake	Skidegate Lk ne		
ats_year	(All)		
Tot Targets	Survey		
	Replicate or potential duplicate (sequential survey dates)		
	1994-08-29	1994-08-30	1994-08-31
	JUVENILE		
	SIMRAD		
	2		
	TARGET94.DAT		
	EYM#20111,DAT#A61400	NA	
Transec	c.c.25%,winds calm,waves 0cm A.T.13.6,H2O 19.8,no moon,Processed by S.T.	c.c.25%,winds calm,waves 0cmA.T.13.6,H2O 19.8,no m Processed by S.T., species proportions reassigned, pr 22Feb95.	Not sure what this is??, pr 22Feb95. This is a revised pop. est. using the correct proportions...S.T.
1	125	125	125
2	459	459	459
3	127	127	127
4	412	412	412
5	107	107	91
6	228	228	228
7	184	184	184
Grand Total	1642	1642	1626

- 94-08-31 appears to be the final but target counts in transect 5 differ. 94-08-29 and 94-08-31 deleted. Kept 94-08-30 since PR re-assigned proportions (i.e., ignore red font colour in table for 94-08-30).

Trapper Lake

lake	Trapper Lk	
ats_year	All	
Sum of targets	Survey Date	
	1991-09-16	1991-09-17
	JUVENILE	
	SIMRAD	
	TARGET91.DAT	
	EYM #20128, "Small" targets.	EYM #20128, "Small"+"faded " targets.
Transect	processed by pr, 29May92	processed by pr 29May92.
1	53	282
2	50	178
3	13	129
4	18	978
5	42	73
Grand Total	176	1640

- 91-09-16 and -17 – different estimates based on target size; unclear which is valid, but the second one seems to be a correction to the first. 91-09-16 deleted.

Tuya Lake

lake	Tuya Lk	
ats_year	All	
Sum of targets	Survey Date	
	1996-09-09	1996-09-10
	JUVENILE	
	SIMRAD	
	TARGET96.DAT	
	ovrcst,lghtrn,lgh tmodwnd,nomn ,AT5-8,ST8.5. 09Sep96/31Oct 96/pr/eym/2011 1/10005/1/0/0/1 2/62/0/29ctds.	ovrcst,lghtrn,lgh tmodwnd,nomn ,AT5-8,ST8.5. Species composition by depth changed.pr(17A pr97).
Transect	processed by pr, 31Oct96.	processed by pr, 31Oct96.
1	265	265
2	284	284
3	284	284
4	348	348
5	421	421
6	31	31
7	227	227
8	224	224
9	0	0
10	22	22
Grand Total	2106	2106

96-09-09 and -10 – 96-09-09 deleted based on updates to species composition in 96-09-10.

Vernon Lake

lake	Vernon Lk			
ats_year	All			
Sum of targets	Survey Date			
	2000-06-27	2000-06-28	2000-09-22	2000-09-23
	JUVENILE			
	SIMRAD			
	TARGET00.DAT			
	27Jun00/27Jul0 0/BH/EYM/2006 9/0/0/0/5/150/0 EY500 replicated these transects.	27Jun00/27Jul0 0/BH/EYM/2006 9/0/0/0/5/150/0 EYM replicated these transects. EY500 @-60dB.	EYM20069 Estimate 22Sep00/20Dec0 0/BH/EYM/20069/ 0/0/0/1/57/0	22Sep00/16Jan01 /BH/EY500/0/0/0/ 1/57/0
Transect	Poor Survey Fish Clumped and on Shore.	Poor Survey Fish Clumped and on Shore.	Hanslit Survey	Hanslit Survey
1	117	90	176	139
2	205	241	311	228
3	522	567	526	454
4	476	520	300	267
Grand Total	1320	1418	1313	1088

- 2000-06-27/28 – different software (?) : EY500 vs EYM? No changes advised
- 2000-09-22/23 – different software (?) : EY500 vs EYM? No changes advised

Woss Lake

lake	Woss Lk			
ats_year	(All)			
Tot Targets	Survey			
	Replicate or potential duplicate (sequential survey dates)			
	1993-11-18	1993-11-19	2000-06-26	2000-06-27
	JUVENILE			
	SIMRAD			
	2			
	TARGET93.DAT		TARGET00.DAT	
	Counted Dec.6/93 SM		NA	
Transec	ovcst, rain,clm,0moon,AT 5.0 WT 11.5. EY-M 20111 gain 10, log 40, dynaline ON.	ovcst,rain, clm, AT 5.0 WT 11.5.	Fair 26Jun00/27Jul00/EY M/BH/20069/0/0/0/ 5/138/0	Fair 26Jun00/27Jul00/EY 500@- 60dB/BH/20069/0/0 /0/5/138/0
1	844	1133	293	0
2	1071	1330	115	0
3	966	1275	30	58
4	511	638	131	86
5	526	529	248	230
Grand Total	3918	4905	817	374

- 93-11-18 and -19 – unclear what the different survey datasets represent; consult STRs (Summary Trip Reports) Neither deleted.
- 2000-06-26/27 – EYM vs EY500 unit differences – no changes advised.

Yakoun Lake

lake	Yakoun Lk	
ats_year	(All)	
Tot Targets	Survey	
	Replicate or potential duplicate (sequential survey dates)	
	1998-08-14	1998-08-15
	ADULT	
	SIMRAD	
	2	
	TARGET98.DAT	
	Fairweather Survey	NA
Transect	Adult Count	Adult Count
1	3	2
2	4	3
3	1	1
4	5	5
5	8	8
6	6	6
7	5	6
8	10	10
9	7	7
10	2	2
11	1	1
12	1	1
Grand Total	53	52

- 98-08-14 and -15 – unclear as to what the difference in targets is due; consult STRs? 98-08-14 deleted.

Other Data Issues

The **Acoustic Target Data Cleanup.R** program checks for a number of other data incongruencies and potential errors, and flags these with a text comment in the **data_issues** field for the affected record(s). Users should review the **data_issues** field, as well as the **sounder_issues**, **acoustic_survey_comments** and **acoustic_survey_notes** fields for issues, prior to using these data in the Target_OUTPUT_data_FINAL worksheet of the Target_OUTPUT_CHK_data.xlsx workbook.

In addition, the ACOUSTIC TARGET DATA CLEANUP LOG.txt file, saved from the RStudio console after program execution, lists the following types of data issues in various fields.

Sounder Issues

Sounder Code Check

The target data from 1977-2009 were mostly generated using two types of echo-location sounders: FURUNO and SIMRAD, with sounder codes 1 = FURUNO; 2 = SIMRAD. Sounder name and sounder code are recorded in the TARGETty.DAT import data on the same line.

Of the 1,324 surveys reviewed, 292 were associated with FURUNO models, 1,026 were done with SIMRAD units, and 3 were associated with BIOSONICS devices, for which no code has been provided. In addition, 3 surveys were identified as FURUNO-based, but with sounder_code = 2 (SIMRAD). It is not clear from the data which component of the sounder information line is incorrect, the code or the model name for these three surveys, and no changes have been made to these records.

sounder_code	sounder_type	n_occurrences	issue
1	FURUNO FM 22	275	
1	FURUNO FM22	8	
1	FURUNO	5	
1	BIOSONICS G12	3	no sounder code assigned
1	FURUNO FE 501	1	
1	FURUNO FG 200	1	
1	FURUNO FG 21	1	
1	FURUNO FM 22	1	
2	SIMRAD	1026	
2	FURUNO FM 22	1	erroneous type or code?
2	FURUNO FE 501	1	erroneous type or code?
2	FURUNO FM22	1	erroneous type or code?
TOTAL		1324	

The apparently erroneous sounder assignments can be filtered for records where sounder_issue contains "Sounder Code/Type discrepancy". The lake surveys and their source files include the following:

sounder_issues	sounder_code	sounder_type	sounder_gain	lake_code	survey_date	source_file	line_number
1 Sounder Code/Type discrepancy;	1	"BIOSONICS G12_T	12	6	1991-07-22	TARGET91.D_T	1809
2 Sounder Code/Type discrepancy;	1	"BIOSONICS G12_T	12	1	1993-02-12	TARGET92.D_T	459
3 Sounder Code/Type discrepancy;	1	"BIOSONICS G12_T	12	23	1998-02-06	TARGET97.D_T	4076
4 Sounder Code/Type discrepancy;	2	"FURUNO FM 2_T	10	29	1980-09-20	TARGET80.D_T	161
5 Sounder Code/Type discrepancy;	2	"FURUNO FE 501_T	9	26	1979-07-27	TARGET79.D_T	676
6 Sounder Code/Type discrepancy;	2	"FURUNO FM22_T	10	26	1980-07-24	TARGET80.D_T	1081

Sounder Gain Check

The sounder gain value is missing (NA) or erroneous (>50) for the following surveys:

sounder_code	sounder_type	sounder_gain	lake_code	survey_date	source_file	line_number
1	"FURUNO FM 22_T	NA	3	2007-01-20	TARGET06.D_T	684
2	"SIMRAD T	NA	6	1988-06-14	TARGET88.D_T	954
2	"SIMRAD T	NA	66	1998-08-06	TARGET98.D_T	1726
2	"SIMRAD T	NA	238	2003-05-26	TARGET03.D_T	1075
2	"SIMRAD T	NA	802	2003-11-25	TARGET03.D_T	891
2	"SIMRAD T	NA	23	2004-08-17	TARGET04.D_T	2171
2	"SIMRAD T	NA	22	2004-10-19	TARGET04.D_T	2826
2	"SIMRAD T	500	241	2005-05-24	TARGET05.D_T	2415
2	"SIMRAD T	NA	3	2005-06-28	TARGET05.D_T	105
2	"SIMRAD T	NA	241	2005-08-02	TARGET05.D_T	2621
2	"SIMRAD T	NA	238	2005-08-05	TARGET05.D_T	1901
2	"SIMRAD T	500	241	2005-09-12	TARGET05.D_T	2724
2	"SIMRAD T	500	238	2005-09-13	TARGET05.D_T	1984
2	"SIMRAD T	500	22	2005-09-18	TARGET05.D_T	3731
2	"SIMRAD T	500	23	2005-09-19	TARGET05.D_T	3427
2	"SIMRAD T	NA	3	2005-09-27	TARGET05.D_T	491
2	"SIMRAD T	500	23	2005-10-04	TARGET05.D_T	3503
2	"SIMRAD T	500	22	2005-10-05	TARGET05.D_T	3898
2	"SIMRAD T	NA	23	2005-11-01	TARGET05.D_T	3579
2	"SIMRAD T	NA	22	2005-11-02	TARGET05.D_T	4065
2	"SIMRAD T	500	241	2005-11-22	TARGET05.D_T	2930
2	"SIMRAD T	500	238	2005-11-28	TARGET05.D_T	2067
2	"SIMRAD T	NA	801	2005-12-07	TARGET05.D_T	1549
2	"SIMRAD T	NA	802	2005-12-07	TARGET05.D_T	1652
2	"SIMRAD T	NA	23	2005-12-12	TARGET05.D_T	3655
2	"SIMRAD T	NA	22	2005-12-14	TARGET05.D_T	4232
2	"SIMRAD T	NA	2	2006-01-04	TARGET05.D_T	3136
2	"SIMRAD T	NA	3	2006-01-09	TARGET05.D_T	684
2	"SIMRAD T	NA	241	2006-01-17	TARGET05.D_T	3033
2	"SIMRAD T	NA	238	2006-01-18	TARGET05.D_T	2150
2	"SIMRAD T	NA	228	2006-01-25	TARGET05.D_T	2233
2	"SIMRAD T	NA	41	2006-02-21	TARGET05.D_T	1386
2	"SIMRAD T	NA	40	2006-02-22	TARGET05.D_T	1050
2	"SIMRAD T	NA	238	2006-05-17	TARGET06.D_T	1585
2	"SIMRAD T	NA	3	2006-06-21	TARGET06.D_T	105
2	"SIMRAD T	NA	23	2006-06-28	TARGET06.D_T	3103
2	"SIMRAD T	NA	3	2006-07-20	TARGET06.D_T	298
2	"SIMRAD T	NA	23	2006-08-22	TARGET06.D_T	3179
2	"SIMRAD T	NA	22	2006-08-23	TARGET06.D_T	3498
2	"SIMRAD T	NA	3	2006-09-19	TARGET06.D_T	491
2	"SIMRAD T	NA	22	2006-11-02	TARGET06.D_T	3665
2	"SIMRAD T	NA	23	2006-11-05	TARGET06.D_T	3255
2	"SIMRAD T	NA	238	2006-11-22	TARGET06.D_T	1668
2	"SIMRAD T	NA	241	2006-11-23	TARGET06.D_T	2763
2	"SIMRAD T	NA	228	2007-02-15	TARGET06.D_T	2244
2	"SIMRAD T	NA	801	2007-02-20	TARGET06.D_T	1399
2	"SIMRAD T	NA	802	2007-02-20	TARGET06.D_T	1502

List of Data Issues Types and their Frequency

The following table is a partial list of the types of data_issues and their frequency in the final output data. Multiple data issues may be identified for a given record. Individual data issues are separated by a semi-colon in the text in the data_issues field. Due to truncation of the text in this table, not all of the data_issues are fully listed below. See the Target_OUTPUT_data_FINAL worksheet of the Target_OUTPUT_CHK_data.xlsx workbook for the complete data_issues information.

data_issues	count
Adj. for size bias. Orig. replicate dated 2000-11-30 omitted;	39
Adj. for size bias. Orig. replicate dated 2001-12-04 omitted;	37
Adj. for size bias. Orig. replicate dated 2004-01-22 omitted;	54
Adj. for size bias. Orig. replicate dated 2004-02-04 omitted;	159
Adj. for size bias. Orig. replicate dated 2008-02-19 omitted;	63
Adj. for size bias. Orig. replicate dated 2008-02-20 omitted;	77
Adj. for size bias. Orig. replicate dated 791219 omitted;	57
Adj. for size bias. Orig. replicates dated 910305 & 910306 omitted;	57
Adjusted data. Orig. replicate dated 860923 omitted;	50
Adjusted data. Orig. replicate dated 930225 omitted;	49
Adjusted data. Orig. replicate dated 930716 omitted;	106
Adjusted data. Orig. replicate dated 950620 omitted;	67
Adjusted for size bias. Other surveys dated 2004-11-29 & 2004-11-30 omitted;	138
Adjusted for size bias. Prev. survey dated 2004-03-02 omitted;	57
Adjusted for size bias. Prev. survey dated 2004-03-09 omitted;	140
Adjusted for size bias. Prev. survey dated 2005-08-04 omitted; Sounder Gain missing or error;	53
Adjusted for size bias. Prev. survey dated 2007-08-23 omitted;	51
Apparent duplicate data date 910916 omitted;	36
Apparent duplicate data dated 2007-02-14 omitted;	62
Apparent duplicate data dated 2007-02-14 omitted; Sounder Gain missing or error;	87
Apparent duplicate data in TARGET90.DAT omitted;	72
Apparent duplicate data in TARGET93.DAT & TARGET94.DAT omitted;	22
Apparent duplicate data in TARGET93.DAT omitted;	179
Apparent duplicate data in TARGET96.DAT omitted;	35
Duplicate dated 2007-02-19 omitted; Sounder Gain missing or error;	101
Missing or invalid survey_date (31-Jun-85), assigned (01-Jul-85) from survey comment info;	38
Missing or invalid survey_date (31-Sep-00), assigned (01-Oct-00) from survey comment info;	26
Missing or invalid survey_date (96/03/--), assigned (960320) from survey comment info;	34
Missing or invalid survey_date (96/03/--), assigned (960322) from survey comment info;	46
NOTE: This lake code (247) may be synonymous with 121 (Nahwitti);	47
NOTE: This lake code (249) may be synonymous with 59 (Phillips);	52
NOTE: This lake code (250) may be synonymous with 50 (Heydon);	53
NOTE: This lake code (251) may be synonymous with 238 (Osyoos (N));	264
NOTE: This lake code (252) may be synonymous with 122 (Quatse);	22
NOTE: This lake code (255) may be synonymous with 241 (Skaha);	139
Near duplicate survey date 860815 incomplete, omitted;	25
Near duplicate survey date 860821 incomplete, omitted;	58
Near duplicate survey date 870421 incomplete, omitted;	43
Near duplicate survey date 910830 incomplete, omitted;	45
Near duplicate with minor diff in targets dated 1998-08-14 omitted;	21
Prop_sockeye and prop_stickleback set to NA because targets == 0;	56729
Prop_sockeye and prop_stickleback set to NA because targets == 0; Sounder Code/Type discrepancy;	275
Prop_sockeye and prop_stickleback set to NA because targets == 0; Sounder Gain missing or error;	2327
Prop_sockeye and prop_stickleback set to NA because targets == 0; Adj. for size bias. Orig. replicate dated 2 _T	39
Prop_sockeye and prop_stickleback set to NA because targets == 0; Adj. for size bias. Orig. replicate dated 2 _T	41
Prop_sockeye and prop_stickleback set to NA because targets == 0; Adj. for size bias. Orig. replicate dated 2 _T	24
Prop_sockeye and prop_stickleback set to NA because targets == 0; Adj. for size bias. Orig. replicate dated 2 _T	114
Prop_sockeye and prop_stickleback set to NA because targets == 0; Adj. for size bias. Orig. replicate dated 2 _T	41

Cont'd.

data issues										count
Prop_sockeye and prop_stickleback set to NA because targets == 0; Adj. for size bias. Orig. replicate dated 2	T									92
Prop_sockeye and prop_stickleback set to NA because targets == 0; Adj. for size bias. Orig. replicate dated 7	T									21
Prop_sockeye and prop_stickleback set to NA because targets == 0; Adj. for size bias. Orig. replicates dated 2	T									21
Prop_sockeye and prop_stickleback set to NA because targets == 0; Adjusted data. Orig. replicate dated 860923	T									130
Prop_sockeye and prop_stickleback set to NA because targets == 0; Adjusted data. Orig. replicate dated 930225	T									131
Prop_sockeye and prop_stickleback set to NA because targets == 0; Adjusted data. Orig. replicate dated 930716	T									74
Prop_sockeye and prop_stickleback set to NA because targets == 0; Adjusted data. Orig. replicate dated 950620	T									103
Prop_sockeye and prop_stickleback set to NA because targets == 0; Adjusted for size bias. Other surveys dated	T									42
Prop_sockeye and prop_stickleback set to NA because targets == 0; Adjusted for size bias. Prev. survey dated	T									12
Prop_sockeye and prop_stickleback set to NA because targets == 0; Adjusted for size bias. Prev. survey dated	T									40
Prop_sockeye and prop_stickleback set to NA because targets == 0; Adjusted for size bias. Prev. survey dated	T									17
Prop_sockeye and prop_stickleback set to NA because targets == 0; Adjusted for size bias. Prev. survey dated	T									129
Prop_sockeye and prop_stickleback set to NA because targets == 0; Apparent duplicate data date 910916 omitted;	T									9
Prop_sockeye and prop_stickleback set to NA because targets == 0; Apparent duplicate data dated 2007-02-14 omi	T									42
Prop_sockeye and prop_stickleback set to NA because targets == 0; Apparent duplicate data dated 2007-02-14 omi	T									82
Prop_sockeye and prop_stickleback set to NA because targets == 0; Apparent duplicate data in TARGET90.DAT omi	T									27
Prop_sockeye and prop_stickleback set to NA because targets == 0; Apparent duplicate data in TARGET93.DAT & T	T									23
Prop_sockeye and prop_stickleback set to NA because targets == 0; Apparent duplicate data in TARGET93.DAT omi	T									127
Prop_sockeye and prop_stickleback set to NA because targets == 0; Apparent duplicate data in TARGET96.DAT omi	T									109
Prop_sockeye and prop_stickleback set to NA because targets == 0; Duplicate dated 2007-02-19 omitted; Sounde	T									53
Prop_sockeye and prop_stickleback set to NA because targets == 0; Missing or invalid survey_date (31-Jun-85),	T									122
Prop_sockeye and prop_stickleback set to NA because targets == 0; Missing or invalid survey_date (31-Sep-00),	T									33
Prop_sockeye and prop_stickleback set to NA because targets == 0; Missing or invalid survey_date (96/03/--),	T									12
Prop_sockeye and prop_stickleback set to NA because targets == 0; Missing or invalid survey_date (96/03/--),	T									34
Prop_sockeye and prop_stickleback set to NA because targets == 0; NOTE: This lake code (247) may be synonymou	T									15
Prop_sockeye and prop_stickleback set to NA because targets == 0; NOTE: This lake code (249) may be synonymou	T									12
Prop_sockeye and prop_stickleback set to NA because targets == 0; NOTE: This lake code (250) may be synonymou	T									11
Prop_sockeye and prop_stickleback set to NA because targets == 0; NOTE: This lake code (251) may be synonymou	T									128
Prop_sockeye and prop_stickleback set to NA because targets == 0; NOTE: This lake code (252) may be synonymou	T									11
Prop_sockeye and prop_stickleback set to NA because targets == 0; NOTE: This lake code (255) may be synonymou	T									149
Prop_sockeye and prop_stickleback set to NA because targets == 0; Near duplicate survey date 860815 incomple	T									17
Prop_sockeye and prop_stickleback set to NA because targets == 0; Near duplicate survey date 860821 incomple	T									14
Prop_sockeye and prop_stickleback set to NA because targets == 0; Near duplicate survey date 870421 incomple	T									56
Prop_sockeye and prop_stickleback set to NA because targets == 0; Near duplicate survey date 910830 incomple	T									9
Prop_sockeye and prop_stickleback set to NA because targets == 0; Near duplicate with minor diff in targets d	T									99
Prop_sockeye and prop_stickleback set to NA because targets == 0; Replicate with different spp comp. dated 19	T									20
Prop_sockeye and prop_stickleback set to NA because targets == 0; Replicate with different spp comp. dated 19	T									38
Prop_sockeye and prop_stickleback set to NA because targets == 0; Replicate with different spp comp. dated 20	T									38
Prop_sockeye and prop_stickleback set to NA because targets == 0; Replicate with unadjusted spp comp. dated 2	T									57
Prop_sockeye and prop_stickleback set to NA because targets == 0; Replicate with unadjusted spp comp. dated 2	T									29
Prop_sockeye and prop_stickleback set to NA because targets == 0; Replicate with unadjusted spp comp. dated 2	T									44
Prop_sockeye and prop_stickleback set to NA because targets == 0; Replicate with unadjusted spp comp. dated 8	T									9
Prop_sockeye and prop_stickleback set to NA because targets == 0; Replicate with unadjusted spp comp. dated 9	T									19
Prop_sockeye and prop_stickleback set to NA because targets == 0; Replicate with unadjusted spp comp. dated 9	T									19
Prop_sockeye and prop_stickleback set to NA because targets == 0; Replicates with different targets & spp com	T									11
Replicate with different spp comp. dated 1995-11-20 omitted;										100
Replicate with different spp comp. dated 1996-09-09 omitted;										41
Replicate with different spp comp. dated 2008-09-16 omitted;										74
Replicate with unadjusted spp comp. dated 2003-09-07 omitted;										31
Replicate with unadjusted spp comp. dated 2007-11-07 omitted;										119
Replicate with unadjusted spp comp. dated 2008-02-22 omitted;										110
Replicate with unadjusted spp comp. dated 860415 omitted;										20
Replicate with unadjusted spp comp. dated 940907 omitted;										21
Replicate with unadjusted spp comp. dated 981202 omitted;										59
Replicates with different targets & spp comp. dated 940829 & 940831 omitted;										22
Revised proportions (was 0.23 sox, 0.77 & 0.08 stix) based on TARGET02ADJ.DAT file;										12
Revised proportions (was 0.23 sox, 0.77 stix) based on TARGET02ADJ.DAT file;										6
Revised proportions (was 0.40 sox, 0.60 stix) based on TARGET05ADJ.DAT file;										6
Revised proportions (was 0.73 sox, 0.27 stix) based on TARGET05ADJ.DAT file;										12
Revised proportions (was 0.83 sox, 0.17 stix) based on TARGET02ADJ.DAT file;										6
Revised proportions (was 0.99 sox, 0.01 stix) based on TARGET02ADJ.DAT file;										12

Cont'd.

data issues	count
Sounder Code/Type discrepancy;	188
Sounder Gain missing or error;	2537
Source_file/survey_date mis-match;	1800
Source_file/survey_date mis-match; Data erroneously stored in TARGET95.DAT, ats_year updated from survey_date	29
Source_file/survey_date mis-match; Prop_sockeye and prop_stickleback set to NA because targets == 0;	1962
Source_file/survey_date mis-match; Prop_sockeye and prop_stickleback set to NA because targets == 0; Data err	115
Source_file/survey_date mis-match; Prop_sockeye and prop_stickleback set to NA because targets == 0; Suspicio	100
Source_file/survey_date mis-match; Species proportions do not add to 1;	1
Source_file/survey_date mis-match; Suspicious replicate of KCA surveys in TARGET84.DAT; may be erroneous lake	60
Source_file/survey_date mis-match; Transect Length = 0 but #Targets > 0;	22
Species proportions do not add to 1;	640
Species proportions do not add to 1; Sounder Gain missing or error;	2
Species proportions do not add to 1; Adjusted data. Orig. replicate dated 950620 omitted;	10
Species proportions do not add to 1; Apparent duplicate data in TARGET93.DAT omitted;	1
Species proportions do not add to 1; Missing or invalid survey_date (31-Sep-00), assigned (01-Oct-00) from su	1
Species proportions do not add to 1; Targets < 0 (set to NA);	1
Species proportions do not add to 1; Transect Length = 0 but #Targets > 0;	13
Targets < 0 (set to NA);	1
Transect Length = 0 but #Targets > 0;	451
Transect Length = 0 but #Targets > 0; Sounder Gain missing or error;	9
Transect Length = 0 but #Targets > 0; Adjusted for size bias. Prev. survey dated 2004-03-02 omitted;	1
Transect Length = 0 but #Targets > 0; Apparent duplicate data in TARGET93.DAT omitted;	7
Transect Length = 0 but #Targets > 0; Duplicate dated 2007-02-19 omitted; Sounder Gain missing or error;	6
Transect Length = 0 but #Targets > 0; Missing or invalid survey_date (96/03/--), assigned (960320) from surve	2
Transect Length = 0 but #Targets > 0; NOTE: This lake code (247) may be synonymous with 121 (Nahwitti);	2
Transect Length = 0 but #Targets > 0; NOTE: This lake code (252) may be synonymous with 122 (Quatse);	2
Transect Length = 0 but #Targets > 0; NOTE: This lake code (255) may be synonymous with 241 (Skaha);	12
Transect Length = 0 but #Targets > 0; Replicate with different spp comp. dated 1995-11-20 omitted;	1
Transect Length = 0 but #Targets > 0; Replicate with different spp comp. dated 1996-09-09 omitted;	1
Transect Length = 0 but #Targets > 0; Replicate with unadjusted spp comp. dated 2003-09-07 omitted;	2
Transect Length = 0 but #Targets > 0; Replicate with unadjusted spp comp. dated 2007-11-07 omitted;	12
Transect Length = 0 but #Targets > 0; Replicate with unadjusted spp comp. dated 2008-02-22 omitted;	6
Transect Length = 0 but #Targets > 0; Replicate with unadjusted spp comp. dated 860415 omitted;	1
Transect Length = 0 but #Targets > 0; Replicates with different targets & spp comp. dated 940829 & 940831 omi	2
TOTAL RECORDS WITH DATA ISSUES ----->	73292